

Trust-Aware Resilient Distributed Observer Design for Connected Vehicle Platoons

Q.H. Nguyen, S. Meng, H. Rafaralahy, M. Haddad, and A. Zemouche

Abstract—Connected vehicle platoons require distributed state estimates, but vehicle-to-vehicle packets may contain falsified states, replayed or delayed data, packet losses, or faulty-neighbor estimates. This paper develops a trust-aware resilient distributed observer for vehicles modeled by a discrete-time kinematic bicycle model. Received data are checked for validity and consistency, and local and distributed consistency scores are mapped to trust-adaptive observer weights. The observer keeps a local-observer anchor while selecting time-varying trusted neighbors, reducing the influence of unreliable sources without discarding cooperative information that remains consistent. We give checkable sufficient conditions for local nonlinear input-to-state stability under bounded model uncertainty, input mismatch, accepted attack values, nonlinear remainders, and trust-induced switching; both direct-anchor and sparse accepted-graph cases are covered. To address detection delay, a finite-window rollback procedure replays recent observer updates after a trust decision changes. Numerical tests on a five-vehicle platoon under position, speed, and packet-drop attacks show that the trust mechanism detects and suppresses corrupted sources while keeping estimation errors bounded. QCar/LIMO platform results show that rollback reduces position error after delayed detection, and the tested trust-modulated adaptive and cooperative adaptive cruise-control scheme preserves positive path-projected spacing during the attack interval.

Index Terms—Connected and Autonomous Vehicles, system state estimation, methods for security and privacy, distributed observers, trust management.

I. INTRODUCTION

AUTONOMOUS road vehicles are no longer only mechanical platforms with added electronics. Advances in onboard sensing, embedded computation, and AI-supported perception are turning them into mobile cyber-physical systems that process large sensor streams and close control loops in real time. However, single-vehicle autonomy is only one part of the emerging mobility architecture. The next step is cooperation: through vehicle-to-vehicle (V2V) and broader vehicle-to-everything (V2X) communication, multiple vehicles can share state information and coordinate maneuvers, enabling platooning, smoother traffic flow, and safer mobility. In

a connected and automated vehicle platoon, each vehicle must estimate not only its own state but also the states of surrounding platoon members from a mixture of onboard sensing and communicated information. This distributed estimation layer is a critical component of cooperative adaptive cruise control and other platoon-level decision loops.

The same connectivity that enables cooperation also enlarges the attack surface. From a systems viewpoint, threats may enter through three main layers: the sensing and perception layer, where spoofed or degraded measurements can mislead the ego vehicle; the actuation and control layer, where manipulated control loops can create unsafe physical motion; and the communication layer, where denial-of-service, replay, delay, or false-data-injection attacks can corrupt the information exchanged between vehicles. The present paper focuses on this communication-driven vulnerability, while also treating faulty neighbor sensing or local estimation as unreliable reported data. This is not only a theoretical concern: cyberattacks and misbehavior in connected automated platoons and Internet-of-Vehicles networks have become practical research issues in intelligent transportation systems [?], [?], [?].

Once compromised information is accepted by a distributed observer, the resulting estimation error can propagate across the communication graph and degrade both safety and closed-loop performance. Conventional distributed observers often assume that exchanged data are reliable, which is restrictive for connected-vehicle applications exposed to cyberattacks, packet drops, and faulty nodes. A resilient observer should therefore decide online which neighbor information remains useful, reduce the influence of unreliable packets, and still preserve cooperation when the received data are consistent with the vehicle motion.

This paper develops a trust-aware distributed observer for resilient platoon state estimation. The main idea is to evaluate the behavioral consistency of received data, convert this consistency into trust scores, and use those scores to adapt the observer weights online, so that the observer reduces the influence of unreliable neighbors while preserving useful cooperation from reliable vehicles.

A preliminary version of this work was accepted at the 2026 IFAC World Congress [?], where the platoon was described by a linear longitudinal model and the switching topology was handled through a joint-connectivity condition. The present paper keeps that trust-based idea but recasts it for a nonlinear platoon model with a resilient stability analysis, delayed-decision recovery, and hardware validation. Its main contributions, stated as advances over that preliminary version, are as follows:

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QCar/LIMO implementation resources are available at https://github.com/kslhuy/QCar2_Cran; the demonstration video is available at <https://youtu.be/t6-soznp6ks>.

- A nonlinear resilient distributed state-estimation architecture for connected vehicle platoons under adversarial communication, based on a discrete-time kinematic bicycle model rather than the linear longitudinal model of the conference version, with the analysis carried out on the original nonlinear error dynamics.
- A behavioral trust framework that combines local consistency, distributed-estimate consistency, freshness, and temporal smoothing to construct a trusted-neighbor set, together with a receiving-side validity gate that turns any accepted corruption into a bounded input for the stability analysis.
- A trust-adaptive observer-weight design with admissible mixing coefficients and local nonlinear input-to-state stability (ISS) conditions for bounded modeling, input, attack, and nonlinear mismatch terms, now established through a sequentially rooted accepted-graph condition that yields direct-anchor (one-step) and sparse-graph (windowed block) contraction results, in place of the joint-connectivity condition used previously.
- A finite-window rollback mechanism, stated as an implementable replay procedure, that re-propagates recent observer updates after delayed trust decisions and is shown to preserve the ISS bound under a dwell-time condition.
- An experimental and closed-loop validation: the observer is implemented on a QCar/LIMO platform under real packet timing, and a trust-modulated ACC/CACC controller is checked for positive path-projected spacing during the attack interval, in addition to the numerical campaign under position, velocity, and packet-drop attacks.

The remainder of the paper is organized as follows. Section ?? reviews related work and motivates the proposed approach, with the positioning relative to representative resilient platoon and state-estimation methods summarized in Table ?. Section ?? introduces the graph notation, vehicle model, and threat model. Section ?? presents the distributed observer architecture. Section ?? defines the trust framework. Section ?? develops the trust-based weight design and stability analysis. Section ?? presents the delayed-decision rollback module. Section ?? reports the simulation study, QCar/LIMO platform implementation, and controller spacing check. Section ?? discusses limitations, and concludes the paper.

II. RELATED WORK AND MOTIVATION

Security and resilience for connected vehicle platoons have been studied through attack detection, fault-tolerant estimation, and mitigation-oriented control. Observer-based methods compare model-predicted states with measurements or received data to detect abnormal behavior [?], [?], [?], [?]. These methods are attractive because they are model-aware and can be implemented locally, but their performance degrades when the observer itself uses corrupted neighbor information.

Consensus and distributed-estimation methods exploit redundancy across multiple agents or sensors [?], [?], [?]. They can improve robustness by cross-validating information,

but many approaches implicitly require sufficiently reliable neighbors or fixed communication weights. Under coordinated attacks, persistent packet loss, or biased local observers, fixed consensus weights can spread wrong estimates instead of isolating them.

Formal secure and Byzantine-resilient estimation provides worst-case guarantees under sparse attacked sensors or bounded adversary fractions [?], [?], [?]. The observability of continuous-time linear systems under a bounded number of arbitrarily corrupted sensor outputs was characterized, along with finite-window and switched-observer reconstruction methods in [?]. Sparse sensor attacks with bounded noise were considered in [?], leading to ℓ_0 - and ℓ_1 -based estimators with analytic error bounds proportional to the noise magnitude. These results provided formal attacked-sensor sparsity, adversary-count, or reconstruction guarantees that are stronger than the conditional ISS claim made here. In contrast, the present objective is to adapt distributed V2V weights online for a nonlinear platoon observer and to connect the estimate to delayed-decision rollback and closed-loop use. More general secure-estimation methods also include nonlinear bank-of-observers selection under sparse sensor attacks [?] and distributed ℓ_1 state-and-fault estimation for multi-agent systems [?]; these provide useful resilient-estimation baselines, but they do not directly provide trust-adaptive V2V observer weights for platoons.

Set-membership and data-driven techniques provide another line of resilient estimation and detection. Zonotopic set-based estimators propagate all states consistent with bounded uncertainties and can guarantee true-state containment under time-varying sensor attacks when sufficient uncompromised sensing remains [?]. Data-driven attack-detection methods instead use attack-free input-output trajectories to identify uncompromised sensors online, including extensions for replay and delay attacks [?]. These methods are useful for guaranteed or model-learning-based security analysis, but they are generally centralized or detection-oriented and do not directly adapt the distributed V2V observer weights used by a platoon controller.

Trust-based approaches assign reliability scores to agents or data streams. Physical-layer and connectivity-based methods use signal fingerprints or communication characteristics to authenticate neighbors [?], [?], while behavior-based methods compare exchanged data with local sensing or expected motion [?], [?]. Context-aware IoV trust management further incorporates application context, rational trust-attribute weighting, and adaptive misbehavior thresholds to improve attack resistance in dynamic vehicular networks [?]. Trust can also be injected into control policies to reduce the influence of unreliable vehicles [?], [?]. However, these trust mechanisms are usually diagnostic, reputation-oriented, or controller-side filters; they do not directly provide a trust-adaptive distributed observer with stability guarantees and delayed-decision rollback.

The present work is motivated by these gaps. Instead of treating trust as an external diagnostic signal, the proposed method feeds trust directly into the observer-weight design. The observer therefore adapts its communication graph online while preserving a local-observer anchor, which improves

recoverability after transient attacks and supports a stability analysis for bounded disturbances. The contribution is therefore incremental and architectural: it couples a behavior-based trust layer with admissible observer weights and sufficient ISS conditions, rather than replacing formal sparse-adversary or Byzantine-resilient estimators.

Notations: For vectors $z_1, \dots, z_N$, $\text{col}(z_1, \dots, z_N)$ denotes their vertical concatenation, and $\varrho(M)$ denotes the spectral radius of a square matrix M . $\mathbf{1}_N$ is the all-ones vector.

III. PROBLEM FORMULATION

In this section, a group of connected automated vehicles traveling on a two-dimensional plane are considered. Each vehicle can measure its own state through onboard sensors and exchange state information with neighboring vehicles via V2V communication. Graph theory is used to characterize the communication topology among vehicles, while a discrete-time kinematic bicycle model is adopted to describe the motion dynamics of each vehicle. Considering potential threats such as data falsification, replay attacks, communication-delay attacks, denial-of-service (DoS) attacks, and erroneous neighbor state reports caused by sensor faults, we aim to design a resilient distributed state observer that enables each vehicle to estimate the state of the entire platoon under unreliable communication while ensuring bounded estimation errors.

A. Kinematic bicycle vehicle model with V2V communication

To describe the dynamics of each vehicle, we use a discrete-time kinematic bicycle model. The state and input of vehicle i are defined as

$$x_i = [X_i \ Y_i \ \psi_i \ v_i]^\top, \quad u_i = [a_i \ \delta_i]^\top,$$

where (X_i, Y_i) is the planar position, ψ_i is the yaw angle, v_i is the longitudinal speed, a_i is the acceleration, and δ_i is the front steering angle. With sampling time T_s and wheelbase L_w , the nominal forward-Euler model is

$$\begin{aligned} x_i(t+1) &= f_b(x_i(t), u_i(t)) + \Delta_i(t), \\ y_i(t) &= h_i(x_i(t)), \end{aligned} \quad (1)$$

where $\Delta_i(t)$ collects modeling mismatch, discretization error, and external disturbances, and

$$f_b(x_i, u_i) = \begin{bmatrix} X_i + T_s v_i \cos \psi_i \\ Y_i + T_s v_i \sin \psi_i \\ \psi_i + \frac{T_s v_i}{L_w} \tan \delta_i \\ v_i + T_s a_i \end{bmatrix}. \quad (2)$$

For the full-state ego-measurement case used in the observer design, $h_i(x_i) = x_i$. If a reduced sensor suite is used, h_i denotes the corresponding local measurement map. Throughout the observer and analysis, state differences are denoted by ordinary subtraction. For the yaw component, the difference is interpreted in local angle coordinates and wrapped with $\text{wrap}_\pi(\cdot)$ when needed.

Let $x = \text{col}(x_1, \dots, x_N)$, $u = \text{col}(u_1, \dots, u_N)$, $\Delta = \text{col}(\Delta_1, \dots, \Delta_N)$, and $f_N(x, u) = \text{col}(f_b(x_1, u_1), \dots, f_b(x_N, u_N))$. The overall vehicle dynamics are

$$\begin{cases} x(t+1) = f_N(x(t), u(t)) + \Delta(t), \\ y_i(t) = h_i(x_i(t)), \end{cases} \quad (3)$$

B. Communication graph

The V2V communication layer is described by a graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \mathcal{A}\}$, where $\mathcal{V} = \{1, 2, \dots, N\}$ is the set of vehicles, $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the edge set describing the communication topology, and $\mathcal{A} = [a_{ij}] \in \mathbb{R}^{N \times N}$ is the adjacency matrix. The neighbor set of vehicle i is

$$\mathcal{N}_i = \{j \in \mathcal{V} \mid (j, i) \in \mathcal{E}\}. \quad (4)$$

The Laplacian matrix $\mathcal{L} = [l_{ij}]$ is defined by

$$l_{ij} = \begin{cases} \sum_{k=1}^N a_{ik}, & i = j, \\ -a_{ij}, & i \neq j. \end{cases} \quad (5)$$

Although the communication layer may be undirected, each vehicle can independently adjust its trust-based communication weights toward neighboring vehicles under potential

TABLE I
POSITIONING OF THE PROPOSED METHOD RELATIVE TO RELATED RESILIENT PLATOON/STATE-ESTIMATION APPROACHES.

Approach	Typical attack class	Dist. observer	Adaptive trust weights	ISS/stability	Formal f -resil.	Rollback	Closed-loop use
Observer residual detection [?], [?], [?], [?]	Sensor/communication anomalies	Partial	No	Case dependent	No	No	Partial
Consensus/distributed estimation [?], [?], [?]	Noise and model mismatch	Yes	No	Yes	No	No	Limited
Formal secure/resilient estimation [?], [?], [?], [?], [?]	Sparse, Byzantine, stealthy attacks	Partial	No	Worst-case	Yes	No	No
Bank-of-observers secure estimation [?]	Sparse sensor attacks	No (bank)	No	Yes	Sparse	No	No
Distributed ℓ_1 state/fault estimation [?]	Sparse faulty agents	Yes	No	Error bound	Sparse	No	No
Set-membership/zonotopic estimation [?]	Time-varying sensor attacks	No	No	Set inclusion	Cond.	No	No
Data-driven attack detection [?]	Sensor, replay, delay attacks	No	No	Feasibility cond.	Cond.	No	No
Trust/context-aware reputation filtering [?], [?], [?], [?], [?], [?]	Misbehavior and unreliable links	Partial	Partial	Limited	No	No	Partial
Proposed trust-aware observer	Bias, intermittent faults, replay/delay, packet drops	Yes	Yes	Conditional ISS	No	Optional	Tested

attacks. Because such trust weights are generally asymmetric, the effective information flow becomes directional. Thus, the trust-weighted virtual communication graph can be modeled as a directed weighted graph. More details about the trust framework and its influence on the observer weights are provided in Section ??.

C. Problem formulation and threat model

Each vehicle estimates the full platoon state using onboard measurements and packets received from neighboring vehicles. The objective is to ensure that the estimation error remains bounded and informative for platoon control even when a subset of received packets is unreliable. The proposed method does not require the host to know which channel is attacked or when the attack begins. It infers packet reliability online using physical consistency, agreement among distributed estimates, packet freshness, and trust smoothing.

We consider four communication and data-integrity faults:

- (1) **FDI**: False-data injection, or data falsification, corrupts communicated state information by adding biases or substituting arbitrary values in selected fields, such as planar position, speed, heading, acceleration, or steering. Such packets may remain syntactically valid but violate physical consistency and intervehicle agreement.
- (2) **Replay/Delay**: Replay attacks resend previously valid packets, whereas delay attacks deliver otherwise valid packets with excessive latency. In both cases, the received information is stale and may pass basic format checks while contradicting the current motion evolution and heading consistency.
- (3) **DoS**: Denial-of-service attacks prevent timely packet delivery through packet dropping, channel jamming, or communication-medium saturation. Their effect is intermittent or persistent packet loss, including extended communication blackouts.
- (4) **Benign Neighbor Fault**: Non-adversarial faults, such as sensor malfunction, calibration drift, or local-estimator divergence, can also produce physically inconsistent neighbor estimates. Although not malicious, these faults can mimic integrity attacks at the packet level.

In the analysis, these effects are modeled as bounded disturbance terms, input mismatch terms, and linearization remainder terms, provided that the corresponding magnitudes are bounded. Packet losses are addressed through packet freshness and the selection of trusted neighbors.

To make this bounded-input interpretation precise, every received packet is first passed through authentication, timestamp, and physical-plausibility gates before it can enter the observer. Denote by $z_l(t)$ the packet content transmitted by vehicle l and by $a_{i,l}(t)$ an additive corruption seen by host i . The value made available to the trust and observer layers is

$$\tilde{z}_{i,l}(t) = \Pi_{\Omega_z}(z_l(t - \tau_{i,l}(t)) + a_{i,l}(t)), \quad (6)$$

when the packet is authenticated and fresh; otherwise it is rejected and $g_{i,l}^{\text{pkt}}(t) = 0$. Here Π_{Ω_z} denotes projection or clipping onto a compact physically admissible packet set,

including feasible position increments, speeds, headings, accelerations, steering angles, and timestamps. Therefore, an arbitrarily large injected value is either rejected by the validity gate or converted into a bounded accepted value. The ISS result below applies to these bounded accepted corruptions; it does not claim robustness to unbounded values injected directly into the observer without a receiving-side gate.

The attacker is assumed to affect V2V data received from compromised or faulty neighbors, but not the host vehicle's onboard measurement channel or the local bicycle model used by the observer. The method is designed to isolate unreliable incoming information and maintain platoon-level safety; it is not intended to reconstruct the exact true state of a malicious vehicle after that vehicle's data have been rejected unless independent reliable measurements remain available. This limitation is standard for distributed observer that prioritize safe estimation over reconstruction of compromised agents.

IV. DISTRIBUTED STATE OBSERVER ARCHITECTURE

This section introduces the observer architecture; the trust mechanism is detailed in Section ?. Two layers are used: a Local Observer ($\mathcal{LO}$) based on onboard sensors, and a Distributed Observer ($\mathcal{DO}$) that fuses information exchanged through communication links.

The notation distinguishes the physical origin of an estimate from the host that stores it. Vehicle j runs its own local observer $\mathcal{LO}_j$ from onboard measurements and broadcasts the corresponding anchor packet $\hat{x}_0^{(j)}$, together with timestamp and optional input information. It also broadcasts its fleet estimate $\hat{x}_j = \text{col}(\hat{x}_j^{(1)}, \dots, \hat{x}_j^{(N)})$. When host vehicle i uses $\hat{x}_0^{(j)}$ in $\mathcal{DO}_i^{(j)}$, this quantity is the local-observer output generated by target vehicle j : it is onboard information only when $i = j$, and it is a received V2V anchor packet when $i \neq j$. Therefore the anchor is not assumed to be automatically reliable; its availability and use are controlled by the packet-validity indicator $g_{i,j}^{\text{pkt}}(t)$ and the anchor acceptance variable $\chi_{i0}^{(j)}(t)$ defined below.

A. General structure of the observer

The observer is defined by

$$(\mathcal{LO}_i) : \hat{x}_0^{(i)}(t+1) = f_b(\hat{x}_0^{(i)}(t), u_i(t)) + F_i(t)(y_i(t) - h_i(\hat{x}_0^{(i)}(t))) \quad (7a)$$

$$(\mathcal{DO}_i^{(j)}) : \hat{x}_i^{(j)}(t+1) = f_b(\hat{x}_{i,c}^{(j)}(t), \hat{u}_j(t)) \quad (7b)$$

$$\begin{aligned} \hat{x}_{i,c}^{(j)}(t) = & \hat{x}_i^{(j)}(t) + \sum_{l \in \mathcal{N}_i} w_{il}^{(j)}(t)(\hat{x}_l^{(j)}(t) - \hat{x}_i^{(j)}(t)) \\ & + w_{i0}^{(j)}(t)(\hat{x}_0^{(j)}(t) - \hat{x}_i^{(j)}(t)), \end{aligned} \quad (7c)$$

where $\hat{x}_0^{(i)}(t)$ is the local-observer state of vehicle i , $F_i(t)$ is the corresponding time-varying gain, and $\mathcal{N}_i$ is the neighbor set of vehicle i in $\mathcal{G}$. The variable $\hat{x}_i^{(j)}(t)$ is vehicle i 's distributed estimate of vehicle j 's state. The intermediate state $\hat{x}_{i,c}^{(j)}(t)$ applies trust-weighted consensus and anchor corrections before the nonlinear bicycle prediction. The weights

$w_{il}^{(j)}(t)$, $l \in \mathcal{N}_i$, and $w_{i0}^{(j)}(t)$ scale the neighbor-consensus and local-observer anchoring terms.

Remark 1. If a host forms $\hat{u}_j(t)$, from a known controller law or an input estimator for $\hat{a}_j(t)$ and $\hat{\delta}_j(t)$, the mismatch

$$\Delta u_j(t) = u_j(t) - \hat{u}_j(t) \quad (8)$$

will be treated as part of the disturbance vector.

The distributed observer $\mathcal{DO}_i^{(j)}$ estimates the state of vehicle $j \in \mathcal{V}$ from vehicle i . The neighbor-consensus term in (??) aligns $\hat{x}_i^{(j)}(t)$ with estimates received from $l \in \mathcal{N}_i$, while $w_{il}^{(j)}(t)$ sets each neighbor's influence. Since consensus alone may drive all estimates to a common biased value under disturbances or model mismatch, the anchoring term $w_{i0}^{(j)}(t)(\hat{x}_0^{(j)}(t) - \hat{x}_i^{(j)}(t))$ injects vehicle j 's target-local anchor as a reference before the bicycle prediction in (??).

B. Virtual communication graph

Each distributed observer $\mathcal{DO}_i^{(j)}$ needs reference information from the local observer $\mathcal{LO}_j$. To include this reference, we introduce a virtual anchor node "0" associated with the local observer $\mathcal{LO}_j$. At time t , the active weighted virtual graph $\bar{\mathcal{G}}^{(j)}(t) = \{\bar{\mathcal{V}}, \bar{\mathcal{E}}^{(j)}(t)\}$ is

$$\begin{aligned} \bar{\mathcal{V}} &= \mathcal{V} \cup \{0\}, \\ \bar{\mathcal{E}}^{(j)}(t) &= \{(l, i) \mid l \in \mathcal{N}_i, w_{il}^{(j)}(t) > 0\} \\ &\quad \cup \{(0, i) \mid w_{i0}^{(j)}(t) > 0\}. \end{aligned} \quad (9)$$

Let $W^{(j)}(t) = [w_{il}^{(j)}(t)]_{i,l=1}^N$, with $w_{il}^{(j)}(t) = 0$ when $l \notin \mathcal{N}_i$, and let $w_0^{(j)}(t) = \text{col}(w_{10}^{(j)}(t), \dots, w_{N0}^{(j)}(t))$. The corresponding weighted adjacency matrix can be written as

$$\bar{\mathcal{A}}^{(j)}(t) = \begin{bmatrix} 0 & 0 \\ w_0^{(j)}(t) & W^{(j)}(t) \end{bmatrix}. \quad (10)$$

In the error dynamics, the anchor node is not a state to be estimated. We therefore use the grounded Laplacian $\bar{\mathcal{L}}^{(j)}(t) = [\bar{l}_{ik}^{(j)}(t)] \in \mathbb{R}^{N \times N}$ obtained by removing the anchor row and column while retaining the anchor weights in the diagonal:

$$\begin{aligned} \bar{l}_{ii}^{(j)}(t) &= \sum_{l \in \mathcal{N}_i \cup \{0\}} w_{il}^{(j)}(t), \\ \bar{l}_{ik}^{(j)}(t) &= \begin{cases} -w_{ik}^{(j)}(t), & k \neq i, k \in \mathcal{N}_i, \\ 0, & k \neq i, k \notin \mathcal{N}_i. \end{cases} \end{aligned} \quad (11)$$

Thus, $I_N - \bar{\mathcal{L}}^{(j)}(t)$ contains the residual self weight on the diagonal and the neighbor weights off diagonal, while the missing row mass $w_{i0}^{(j)}(t)$ enters separately through the local-anchor input $w_0^{(j)}(t) \otimes A_j(t)$ in the error recursion. This virtual graph combines local estimation through node 0 and neighbor consensus through $\mathcal{G}$, so each local observer can act as an anchor for distributed estimation.

C. Main objectives

Fixed observer weights assume reliable data exchange, which is not valid when compromised vehicles transmit falsified information. The objective is to define a trust measure for received data and use it to update $w_{il}^{(j)}(t)$ and $w_{i0}^{(j)}(t)$ online, while choosing the target-local gain $F_j(t)$ so the local nonlinear bicycle-error synthesis in Section ?? yields an ISS convergence bound. In particular, there exist constants $c_0, c_1 > 0$ and $\lambda \in (0, 1)$ such that

$$\begin{aligned} \|e_0^{(j)}(t)\| &\leq c_0 \lambda^t \|e_0^{(j)}(0)\| + c_0 \bar{d}_0, \\ \|e_{\mathcal{DO}}^{(j)}(t)\| &\leq c_1 \lambda^t (\|e_{\mathcal{DO}}^{(j)}(0)\| + \|e_0^{(j)}(0)\|) + c_1 (\bar{d}_0 + \bar{d}_{\mathcal{DO}}), \end{aligned} \quad (12)$$

where $e_0^{(j)}$ and $e_{\mathcal{DO}}^{(j)}$ denote the local and distributed estimation errors defined in Section ??, and $\bar{d}_0$, $\bar{d}_{\mathcal{DO}}$, and $\bar{d}_{\mathcal{DO}}$ are the bounded-input constants in Assumption ??.

V. TRUST SCORE FRAMEWORK

To support resilient estimation, each vehicle evaluates the reliability of received data. The proposed trust framework quantifies how much a host vehicle i trusts another vehicle l . Its structure is shown in Fig. ?. The formulas in this section define one implementable trust map. The stability result in Section ?? does not rely on a particular statistical optimality property of this map; it requires only that the resulting weights are nonnegative, bounded, and satisfy the stated contraction/connectivity checks. This separation is useful in practice: the trust layer can be replaced by another detector, while the observer analysis remains valid as long as the induced weights satisfy the same admissibility conditions.

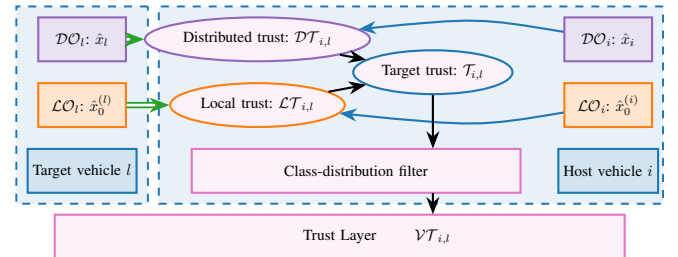


Fig. 1. Trust framework.

In this context, we define the following types of trust:

- **Local Trust** $\mathcal{LT}_{i,l}$: reliability of vehicle l 's local-observer data.
- **Distributed Trust** $\mathcal{DT}_{i,l}$: agreement between distributed estimates from l and those of i .
- **Target Trust** $\mathcal{T}_{i,l}$: instantaneous fusion of Local Trust and Distributed Trust.
- **Vehicle Trust** $\mathcal{V} \mathcal{T}_{i,l}$: class-filtered trust score used by the observer.

Before trust computation, each received V2V packet is checked for authenticity and freshness:

$$g_{i,l}^{\text{pkt}}(t) = \begin{cases} 1, & \text{if the message is authenticated and fresh,} \\ 0, & \text{otherwise.} \end{cases} \quad (13)$$

Only packets with $g_{i,l}^{\text{pkt}}(t) = 1$ are used for trust evaluation.

The following subsections define these scores and their use in the observer.

A. Local trust

Local Trust $\mathcal{LT}_{i,l}$ measures the confidence of host vehicle i in the local-observer output of neighbor $l \in \mathcal{N}_i$. It uses bicycle-state consistency indicators: planar position, speed, heading, and an optional input-consistency check for acceleration and steering. Missing data are handled by a hold-and-decay rule.

1) *Planar position consistency*: Let $p_i = [X_i, Y_i]^\top$. The host transforms its onboard relative measurement of vehicle l into the global frame and denotes it by $r_{i,l}^{\text{meas}}(t) \in \mathbb{R}^2$. The relative position implied by the local observers is

$$r_{i,l}^{\text{obs}}(t) = \begin{bmatrix} \hat{X}_0^{(l)}(t) - \hat{X}_0^{(i)}(t) \\ \hat{Y}_0^{(l)}(t) - \hat{Y}_0^{(i)}(t) \end{bmatrix}.$$

The position-consistency score is

$$\iota_{i,l}^{\text{pos}} = \max \left(1 - \frac{\|r_{i,l}^{\text{obs}} - r_{i,l}^{\text{meas}}\|}{\varepsilon_p + \|r_{i,l}^{\text{meas}}\|}, 0 \right),$$

where $\varepsilon_p > 0$ avoids division by zero.

2) *Speed consistency*: The host compares the relative speed implied by the two local observers with the onboard measured relative speed $v_{i,l}^{\text{rel, meas}}(t)$:

$$\iota_{i,l}^{\text{speed}} = \max \left(1 - \frac{|\hat{v}_0^{(l)}(t) - \hat{v}_0^{(i)}(t) - v_{i,l}^{\text{rel, meas}}|}{\varepsilon_v + |v_{i,l}^{\text{rel, meas}}|}, 0 \right),$$

where $\varepsilon_v > 0$ avoids division by zero.

3) *Heading consistency*: Heading is checked using the reported yaw and a motion-derived yaw estimate. From two consecutive accepted position reports from vehicle l ,

$$\psi_l^{\text{pos}}(t) = \text{atan2}(\hat{Y}_0^{(l)}(t) - \hat{Y}_0^{(l)}(t-1), \hat{X}_0^{(l)}(t) - \hat{X}_0^{(l)}(t-1)).$$

If radar azimuth or LiDAR shape orientation is available, it can replace or complement $\psi_l^{\text{pos}}(t)$. The heading score is

$$\iota_{i,l}^{\text{heading}} = \max \left(1 - \frac{|\text{wrap}_\pi(\hat{\psi}_0^{(l)}(t) - \psi_l^{\text{pos}})|}{\varepsilon_\psi + \psi_{\text{th}}}, 0 \right),$$

where $\psi_{\text{th}} > 0$ is a nominal heading tolerance. This check implements the heading-verification mechanism based on consecutive position reports or bearing measurements.

4) *Optional input consistency*: If acceleration and steering commands are included in the received packet, the host checks them against finite-difference speed and yaw-rate estimates:

$$a_l^{\text{fd}}(t) = \frac{\hat{v}_0^{(l)}(t) - \hat{v}_0^{(l)}(t-n)}{nT_s},$$

$$\omega_l^{\text{bike}}(t) = \frac{\hat{v}_0^{(l)}(t)}{L_w} \tan \hat{\delta}_l(t).$$

With $\omega_l^{\text{fd}}(t) = \text{wrap}_\pi(\hat{\psi}_0^{(l)}(t) - \hat{\psi}_0^{(l)}(t-n))/(nT_s)$,

$$\iota_{i,l}^{\text{input}} = \max \left(1 - \frac{|\hat{a}_l - a_l^{\text{fd}}| + |\omega_l^{\text{bike}} - \omega_l^{\text{fd}}|}{\varepsilon_u + a_{\text{th}} + \omega_{\text{th}}}, 0 \right),$$

where a_{th} and ω_{th} are nominal acceleration and yaw-rate tolerances. If the command channels are not transmitted, $\iota_{i,l}^{\text{input}}$ is omitted from the fusion.

5) *Fusion and update rule*: For simplicity and reproducibility, the available indicators are fused by a weighted geometric product. Let $\omega_p, \omega_v, \omega_\psi, \omega_u \geq 0$ denote channel exponents. When the input channel is available, the local trust score is

$$\mathcal{LT}_{i,l}^{\text{raw}}(t) = \left(\iota_{i,l}^{\text{pos}}(t) \right)^{\omega_p} \left(\iota_{i,l}^{\text{speed}}(t) \right)^{\omega_v} \times \left(\iota_{i,l}^{\text{heading}}(t) \right)^{\omega_\psi} \left(\iota_{i,l}^{\text{input}}(t) \right)^{\omega_u}. \quad (14)$$

If the optional input channel is unavailable, the last factor is omitted. Missing or invalid packets are handled by

$$\mathcal{LT}_{i,l}(t) = \begin{cases} \mathcal{LT}_{i,l}^{\text{raw}}(t), & \text{if } g_{i,l}^{\text{pkt}}(t) = 1, \\ (1 - \lambda_h) \mathcal{LT}_{i,l}(t-1), & \text{otherwise,} \end{cases} \quad (15)$$

where λ_h sets the decay rate during data loss.

Remark 2. Multiplicative fusion is conservative: one inconsistent channel is enough to reduce the local trust score. Larger exponents assign stronger influence to the corresponding consistency channel. The decay λ_h can increase with consecutive missed packets.

B. Distributed trust

Distributed Trust applies the same idea to exchanged distributed estimates. It is defined as

$$\mathcal{DT}_{i,l}(t) = \begin{cases} \gamma_{i,l}^{\text{self}}(t) \cdot \gamma_{i,l}^{\text{local}}(t) \cdot \gamma_{i,l}^{\text{host}}(t) & \text{if } g_{i,l}^{\text{pkt}}(t) = 1 \\ (1 - \lambda_h) \mathcal{DT}_{i,l}(t-1), & \text{otherwise} \end{cases} \quad (16)$$

where $\gamma_{i,l}^{\text{self}}$, $\gamma_{i,l}^{\text{host}}$, and $\gamma_{i,l}^{\text{local}}$ check consistency with the sender's local observer, the host's distributed estimate, and the host's local measurements, respectively.

1) *Sender self-consistency*: The sender's distributed estimate of its own state should remain close to its local observer state:

$$\gamma_{i,l}^{\text{self}}(t) = \exp \left(-(\hat{x}_l^{(l)}(t) - \hat{x}_0^{(l)}(t))^\top \Sigma_{\text{self}}^{-1} \times (\hat{x}_l^{(l)}(t) - \hat{x}_0^{(l)}(t)) \right), \quad (17)$$

where Σ_{self} is the covariance matrix associated with the sender self-consistency residual.

2) *Host-estimate consistency*: The host compares the estimate received from target l with its own estimate using the Mahalanobis distance:

$$\mu_{i,l}^{(j)}(t) = \left(\hat{x}_i^{(j)}(t) - \hat{x}_l^{(j)}(t) \right)^\top \Sigma_{\text{host}}^{-1} \left(\hat{x}_i^{(j)}(t) - \hat{x}_l^{(j)}(t) \right) \quad (18)$$

where Σ_{host} is the diagonal covariance matrix of state-channel discrepancies:

$$\Sigma_{\text{host}} = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2), \quad (19)$$

where n is the number of states and σ_k^2 is the variance of the k th state discrepancy.

The total discrepancy score is

$$D_{i,l}^{\text{host}}(t) = \sum_{j=1}^N \mu_{i,l}^{(j)}(t). \quad (20)$$

A small $D_{i,l}^{\text{host}}(t)$ means that vehicle l 's estimates are close to the host's estimates. The corresponding trust factor is

$$\gamma_{i,l}^{\text{host}}(t) = \exp(-D_{i,l}^{\text{host}}(t)). \quad (21)$$

where $\gamma_{i,l}^{\text{host}}(t) \in (0, 1]$.

3) *Consistency with host's local measurement*: Host vehicle i uses relative measurements to nearby vehicles, e.g., planar relative position, range, bearing, relative speed, or heading difference, to check whether the estimates from target l are consistent with local sensing.

Let $y_{i,j}(t) \in \mathbb{R}^{m_{ij}}$ be the relative measurement from vehicle i to vehicle j , e.g., the global-frame relative position $p_j(t) - p_i(t)$ or a range-bearing vector transformed through the host pose.

The estimate received from target l should approximately satisfy

$$H(\hat{x}_l^{(j)}(t) - \hat{x}_l^{(i)}(t)) \approx y_{i,j}(t).$$

where $H \in \mathbb{R}^{m_{ij} \times n}$ extracts the measured states.

The consistency error is

$$\varepsilon_{i,l}^{(j)}(t) = H(\hat{x}_l^{(j)}(t) - \hat{x}_l^{(i)}(t)) - y_{i,j}(t), \quad (22)$$

Using Σ_{local} as in (??), define

$$E_{i,l}^{(j)}(t) = (\varepsilon_{i,l}^{(j)}(t))^T \Sigma_{\text{local}}^{-1} \varepsilon_{i,l}^{(j)}(t). \quad (23)$$

The total consistency error is

$$E_{i,l}(t) = \sum_{j \in \mathcal{N}_i} E_{i,l}^{(j)}(t). \quad (24)$$

A higher $E_{i,l}(t)$ means that the estimates from target l disagree with local relative measurements.

The trust factor is

$$\gamma_{i,l}^{\text{local}}(t) = \exp(-E_{i,l}(t)), \quad (25)$$

where $\gamma_{i,l}^{\text{local}}(t) \in (0, 1]$.

Remark 3. The covariance matrices Σ_{host} and Σ_{local} can be estimated from a recent nominal-data window at vehicle i , avoiding covariance transmission between nodes.

C. Target trust and temporal evolution

We combine the Local Trust and Distributed Trust into a single instantaneous trust score for target l :

$$\mathcal{T}_{i,l}(t) = \mathcal{L}\mathcal{T}_{i,l}(t) \cdot \mathcal{D}\mathcal{T}_{i,l}(t). \quad (26)$$

$\mathcal{T}_{i,l}(t)$ measures the instantaneous quality of target l 's data. Because this value can vary abruptly, a class-distribution temporal filter is used instead of applying the instantaneous scalar directly. Let $q = 5$ denote the number of ordered trust

classes: *Unreliable*, *Poor*, *Acceptable*, *Good*, and *Excellent*. Define the class scores

$$\kappa_m = \frac{m-1}{q-1}, \quad m = 1, \dots, q, \quad (27)$$

and the corresponding intervals

$$\mathcal{I}_m = \begin{cases} \left[\frac{m-1}{q}, \frac{m}{q} \right), & m = 1, \dots, q-1, \\ \left[\frac{q-1}{q}, 1 \right], & m = q. \end{cases} \quad (28)$$

The instantaneous trust value is discretized into a one-hot class vector

$$r_{i,l,m}(t) = \begin{cases} 1, & \mathcal{T}_{i,l}(t) \in \mathcal{I}_m, \\ 0, & \text{otherwise,} \end{cases} \quad r_{i,l}(t) \in \mathbb{R}^q. \quad (29)$$

For example, $\mathcal{T}_{i,l}(t) = 0.72$ belongs to the *Good* class for $q = 5$, so $r_{i,l}(t) = [0, 0, 0, 1, 0]^T$.

The accumulated class evidence is updated as

$$R_{i,l}(t+1) = (1 - \beta_{\text{trust}} \mathcal{T}_{i,l}(t)) R_{i,l}(t) + r_{i,l}(t), \quad (30)$$

where $R_{i,l}(t) \in \mathbb{R}_{\geq 0}^q$ is initialized as $R_{i,l}(0) = 0$ unless prior interaction evidence is available, and $\beta_{\text{trust}} \in [0, 1]$ controls the decay of past class evidence. The modulation by $\mathcal{T}_{i,l}(t)$ prevents one isolated low-quality sample from immediately erasing the accumulated history, while repeated low-quality samples keep adding evidence to the lower classes. This deliberately asymmetric memory makes trust rejection less sensitive to isolated outliers, but it also delays recovery after a repaired node becomes reliable again; consequently, β_{trust} and the rejection threshold should be calibrated with benign false-positive and recovery data. To regularize the class distribution, let $\alpha_{\text{tr}} \in \mathbb{R}_{\geq 0}^q$ be a prior trust-class distribution satisfying $\sum_{m=1}^q \alpha_{\text{tr},m} = 1$, and let $C_{\text{tr}} > 0$ be a confidence parameter. The prior α_{tr} represents the initial belief about vehicle l before enough packets are observed; if no prior information is available, one can use the uniform choice $\alpha_{\text{tr},m} = 1/q$. The parameter C_{tr} controls how strongly this prior affects the estimate: larger C_{tr} gives more weight to the prior, while smaller C_{tr} lets online evidence dominate faster. The normalized trust-class distribution is

$$S_{i,l,m}(t) = \frac{R_{i,l,m}(t) + C_{\text{tr}} \alpha_{\text{tr},m}}{C_{\text{tr}} + \sum_{s=1}^q R_{i,l,s}(t)}, \quad m = 1, \dots, q. \quad (31)$$

The smoothed vehicle trust used by the observer is the expected trust level

$$\mathcal{V}\mathcal{T}_{i,l}(t) = \sum_{m=1}^q \kappa_m S_{i,l,m}(t). \quad (32)$$

A normalized version of (??)–(??) is equivalent to a finite-class Dirichlet reputation posterior mean with confidence prior $C_{\text{tr}} \alpha_{\text{tr}}$ [?]. A larger β_{trust} gives a more reactive trust history, while a smaller value gives smoother trust evolution. The confidence C_{tr} controls how strongly the prior influences the estimate when only a short interaction history is available. The observer uses $\mathcal{V}\mathcal{T}_{i,l}(t)$ to weigh information received from vehicle l .

D. Practical parameter selection

The trust parameters are selected from nominal data and physical limits rather than tuned directly on attack cases. First, the receiver defines the admissible packet set Ω_z in (??) from vehicle limits, road geometry, timestamp tolerance, maximum speed, acceleration, steering angle, and feasible inter-sample motion. Second, the covariance matrices Σ_{self} , Σ_{host} , and Σ_{local} are estimated from an attack-free calibration window and regularized with positive lower bounds on every diagonal entry. Third, the exponents $\omega_p, \omega_v, \omega_\psi, \omega_u$ encode channel confidence: a channel with lower nominal noise or higher safety relevance receives a larger exponent, while an unavailable channel is removed from the product. Fourth, the threshold $\theta_{\min}$ is chosen from nominal validation data so that benign packets are rarely rejected, then checked under representative faults. Finally, β_{trust} and C_{tr} set the detection–recovery trade-off: larger β_{trust} reacts faster to repeated low-trust evidence, while larger C_{tr} prevents short histories from dominating the prior. After these values are fixed, the observer weights are checked against the stability conditions in Section ??.

VI. TRUST-BASED OBSERVER WEIGHT DESIGN

The trust framework provides each vehicle with smoothed vehicle-trust scores $\mathcal{V}\mathcal{T}_{i,l}(t)$ representing the reliability of information from other vehicles. Fix a rejection threshold $\theta_{\min} \in (0, 1]$. Throughout this section, $\|\cdot\|_*$ denotes a fixed norm on the bicycle state space, and all induced and block norms are formed from this same state norm unless stated otherwise. This section shows how the trust values define adaptive observer weights $w_{il}^{(j)}(t)$ and establishes conditions for stability of the estimation error dynamics.

A. Trust-based weights

Using the virtual graph $\bar{\mathcal{G}}^{(j)}$, each vehicle i first selects the trusted neighbor sources at time t :

$$\mathcal{LN}_i^{(j)}(t) = \left\{ l \in \mathcal{N}_i \mid \mathcal{V}\mathcal{T}_{i,l}(t) \geq \theta_{\min} \right\}. \quad (33)$$

The selected trust scores are then used directly as reliability scores,

$$s_{i,l}^{(j)}(t) = \begin{cases} \mathcal{V}\mathcal{T}_{i,l}(t), & l \in \mathcal{LN}_i^{(j)}(t), \\ 0, & \text{otherwise,} \end{cases} \quad (34)$$

$$S_i^{(j)}(t) = \epsilon + \sum_{r \in \mathcal{N}_i} s_{i,r}^{(j)}(t), \quad (35)$$

where $\epsilon > 0$ avoids division by zero. Let $\eta_0, \eta_N \in [0, 1]$ denote the local-anchor and neighbor-consensus budgets, respectively, with

$$\eta_0 + \eta_N \leq 1. \quad (36)$$

Let $\chi_{i0}^{(j)}(t) \in \{0, 1\}$ indicate that the local-anchor packet $\hat{x}_0^{(j)}(t)$ is available and accepted at host i . For the target's own host, $\chi_{j0}^{(j)}(t) = 1$ and $\mathcal{L}\mathcal{T}_{j,j}(t) = 1$; if the anchor packet is unavailable or rejected, $\chi_{i0}^{(j)}(t) = 0$ and the anchor weight is zero. The local-anchor weight is

$$w_{i0}^{(j)}(t) = \begin{cases} \eta_0 \chi_{i0}^{(j)}(t) \mathcal{L}\mathcal{T}_{i,j}(t), & \chi_{i0}^{(j)}(t) = 1, \\ 0, & \mathcal{L}\mathcal{T}_{i,j}(t) \geq \theta_{\min}, \\ & \text{otherwise,} \end{cases} \quad (37)$$

and the neighbor weights are

$$w_{il}^{(j)}(t) = \eta_N \frac{s_{i,l}^{(j)}(t)}{S_i^{(j)}(t)}, \quad l \in \mathcal{N}_i. \quad (38)$$

Thus, rejected neighbors receive zero weight, while accepted neighbors share at most the budget η_N in proportion to their trust values. The remaining mass is the implicit self-prediction weight

$$w_{is}^{(j)}(t) = 1 - w_{i0}^{(j)}(t) - \sum_{l \in \mathcal{N}_i} w_{il}^{(j)}(t). \quad (39)$$

The following lemma shows that the proposed trust-based weight law always generates admissible mixing coefficients. In particular, the correction step is a nonnegative weighted combination of the self prediction, trusted neighbors, and the local-observer anchor. This property is important because it prevents the observer from extrapolating unreliable information and guarantees that bounded attacks enter the error dynamics only through bounded weighted inputs.

Lemma 1 (Admissibility of trust weights). The weight law (??)–(??) satisfies $w_{il}^{(j)}(t) \geq 0$, $w_{i0}^{(j)}(t) \geq 0$, and

$$\sum_{l \in \mathcal{N}_i \cup \{0\}} w_{il}^{(j)}(t) \leq 1 \quad (40)$$

for all i, j and all t .

Proof. By construction, $s_{i,l}^{(j)}(t) \geq 0$ and $S_i^{(j)}(t) > 0$, hence $w_{il}^{(j)}(t) \geq 0$ for all $l \in \mathcal{N}_i$. Also, $0 \leq w_{i0}^{(j)}(t) \leq \eta_0$ because $\chi_{i0}^{(j)}(t) \in \{0, 1\}$ and $\mathcal{L}\mathcal{T}_{i,j}(t) \in [0, 1]$. The neighbor contribution satisfies

$$\sum_{l \in \mathcal{N}_i} w_{il}^{(j)}(t) = \eta_N \frac{\sum_{l \in \mathcal{N}_i} s_{i,l}^{(j)}(t)}{\epsilon + \sum_{l \in \mathcal{N}_i} s_{i,l}^{(j)}(t)} \leq \eta_N.$$

Therefore,

$$\sum_{l \in \mathcal{N}_i \cup \{0\}} w_{il}^{(j)}(t) = w_{i0}^{(j)}(t) + \sum_{l \in \mathcal{N}_i} w_{il}^{(j)}(t) \leq \eta_0 + \eta_N \leq 1. \quad \blacksquare$$

The lemma provides bounded mixing, but bounded mixing alone is not sufficient for convergence because the bicycle model contains prediction dynamics. The accepted trust-induced weights must also provide enough local-anchor mass, or enough rooted multi-step information flow, to contract the distributed prediction error. Let

$$M^{(j)}(t) = I_N - \bar{\mathcal{L}}^{(j)}(t). \quad (41)$$

By Lemma ??, $M^{(j)}(t)$ is nonnegative and its i th row sum is $1 - w_{i0}^{(j)}(t)$. Therefore, in the infinity norm,

$$\|M^{(j)}(t)\|_\infty = \max_{i \in \mathcal{V}} (1 - w_{i0}^{(j)}(t)). \quad (42)$$

The proposed weight law gives a direct lower bound on this missing row mass whenever the target-local anchor packet is accepted.

Lemma 2 (Anchor-induced one-step contraction). Suppose that, for target j , the accepted anchor packet is available to every host at time t and satisfies $\chi_{i0}^{(j)}(t) = 1$ and

$\mathcal{LT}_{i,j}(t) \geq \theta_{\min}$ for all $i \in \mathcal{V}$. Then the weight law (??) gives

$$w_{i0}^{(j)}(t) \geq \underline{w}_0, \quad \underline{w}_0 := \eta_0 \theta_{\min}. \quad (43)$$

Consequently,

$$\|M^{(j)}(t)\|_{\infty} \leq 1 - \underline{w}_0. \quad (44)$$

If the bicycle map is Lipschitz on the admissible compact region in the same state norm with constant $L_{x,j}$ and

$$L_{x,j}(1 - \underline{w}_0) < 1, \quad (45)$$

then the homogeneous distributed prediction error is a one-step contraction.

Proof. Under the stated acceptance condition, (??) gives $w_{i0}^{(j)}(t) = \eta_0 \mathcal{LT}_{i,j}(t) \geq \eta_0 \theta_{\min}$. The row-sum identity (??) then gives $\|M^{(j)}(t)\|_{\infty} \leq 1 - \underline{w}_0$. Combining this with (??) yields the contraction factor in (??). ■

When direct anchors are intermittent, the same argument must be applied over a window. These weights induce a switching communication graph because the trusted-neighbor set changes online. A plain union spanning tree is not enough unless the anchor information can move through the matrix product in time order. We therefore use the following sequential rooted accepted-graph condition, where an edge $l \rightarrow i$ means that the accepted weight $w_{il}^{(j)}$ is positive.

Assumption 1 (Sequentially rooted accepted graph). *There exist an integer $T_c > 0$ and a scalar $\underline{w} > 0$ such that, for every target j , every start time $t \geq 0$, and every host $i \in \mathcal{V}$, there are nodes $v_0 = 0, v_1, \dots, v_m = i$ and times $t \leq \tau_1 < \dots < \tau_m \leq t + T_c - 1$ for which the accepted edge $v_{r-1} \rightarrow v_r$ is present in $\bar{\mathcal{G}}^{(j)}(\tau_r)$. Each accepted edge weight on these temporal paths is at least $\underline{w}$, and each implicit self-prediction weight used to carry information between two consecutive path times is also at least $\underline{w}$. Moreover, if the path reaches the destination host i at a time $\tau_m < t + T_c - 1$, then the implicit self-prediction weight of host i is also at least $\underline{w}$ at every remaining instant $\tau_m < \tau \leq t + T_c - 1$, so that the anchor-induced row-sum deficit at host i is preserved until the end of the window.*

Assumption ?? means that every host estimate of target j receives anchor information either directly or through trusted intermediate estimates within each window. This is stronger than ordinary connectivity: the root must be the target-local observer node, the links must be accepted by the trust layer, and the path must be ordered along the product. A standard jointly rooted union condition implies this assumption after enlarging the window when accepted links recur in every joint-connectivity interval and the implicit self-prediction weights are uniformly positive; otherwise the product condition below must be enforced by a communication supervisor or checked on each completed window.

Lemma 3 (Rooted block contraction). *If Assumption ?? holds, then the grounded trust-mixing product satisfies*

$$\|M^{(j)}(t + T_c - 1) \cdots M^{(j)}(t)\|_{\infty} \leq 1 - \underline{w}^{T_c}. \quad (46)$$

Let $L_{x,j}(\tau) \leq L_{x,j}$ be a scalar one-step Lipschitz bound that applies uniformly to all hosts estimating target j at time τ . For the separable interleaved prediction-mixing estimate below, define

$$G_{x,j}(T_c) := \sup_{t \geq 0} \prod_{r=0}^{T_c-1} L_{x,j}(t+r) \leq L_{x,j}^{T_c}. \quad (47)$$

When time-resolved bounds are unavailable, the conservative choice is $G_{x,j}(T_c) = L_{x,j}^{T_c}$. The one-step gains are scalar and therefore factor through the mixing matrices. For the componentwise error-size vector

$$E^{(j)}(t) = \text{col} \left(\|e_1^{(j)}(t)\|_*, \dots, \|e_N^{(j)}(t)\|_* \right), \quad (48)$$

the one-step homogeneous recursion satisfies, componentwise,

$$E^{(j)}(t+1) \leq L_{x,j}(t) M^{(j)}(t) E^{(j)}(t). \quad (49)$$

Unrolling this recursion over one block gives

$$E^{(j)}(t + T_c) \leq G_{x,j}(T_c) (M^{(j)}(t + T_c - 1) \cdots M^{(j)}(t)) \times E^{(j)}(t). \quad (50)$$

Thus $G_{x,j}(T_c)$ is an upper bound for the product of the one-step gains in every admissible interleaved block; it is not an independently chosen gain for the prediction-only composition. This explains why the scalar prediction gain and the trust-mixing product separate. If

$$G_{x,j}(T_c)(1 - \underline{w}^{T_c}) \leq \alpha_c < 1, \quad (51)$$

then the distributed homogeneous error is contractive over every window of length T_c . The one-step condition (??) is the special case $T_c = 1$ with direct anchor acceptance at every step.

Proof. The matrices $M^{(j)}(\tau)$ are nonnegative and substochastic. A positive anchor weight creates a strict row-sum deficit in the row receiving that anchor. For any host i , Assumption ?? gives a temporal accepted path from the anchor to i . Each accepted edge transmits at least a factor $\underline{w}$ of the existing deficit to the next node, and each waiting step preserves at least a factor $\underline{w}$ through the implicit self-prediction weight. Since the path transmissions and waiting steps occupy no more than T_c time instants, the terminal-waiting clause of Assumption ?? guarantees that, once the deficit reaches host i at time τ_m , it is carried by a factor of at least $\underline{w}$ per step up to $t + T_c - 1$. Hence it appears in the i th row sum of the full product, which is at most $1 - \underline{w}^{T_c}$. Taking the maximum row sum gives (??). Combining this result with the separable interleaved bound (??) gives (??). ■

Thus, contraction is not assumed from trust scoring alone. The weight law supplies the anchor lower bound through $\eta_0 \theta_{\min}$ when anchor packets are directly accepted, and Assumption ?? supplies the corresponding lower bound for sparse or intermittent trusted paths. If all reliable paths to a target are rejected, the theorem applies only to the reachable part of the platoon, and that target cannot be reconstructed from communication alone.

B. Observer design procedure

The complete design can be implemented through the following steps.

- 1) Select a compact operating region $\Omega_x \times \Omega_u$ from vehicle limits and the receiving-side gate (??). Compute conservative bounds for the bicycle Lipschitz constants $L_{x,j}$ and $L_{u,j}$ on this region.
- 2) Choose the local observer gain $F_j(t)$ so that the zero-input local error recursion is uniformly locally exponentially stable. With full-state local measurements, verify the resulting linear time-varying (LTV) local-error recursion along the considered trajectory using the parameter-dependent Lyapunov inequality (??) in Lemma ?? **Frozen-time eigenvalue placement alone is not sufficient in the time-varying case.**
- 3) Fix the trust parameters using the nominal-data procedure in Section ??, then compute $\mathcal{VT}_{i,l}(t)$, $\mathcal{LT}_{i,j}(t)$, and the acceptance variables $g_{i,l}^{\text{pkt}}(t)$ and $\chi_{i0}^{(j)}(t)$ online.
- 4) Compute the anchor and neighbor weights from (??)–(??).
- 5) Check the contraction condition (??) or (??). If the check fails for a target, the stability certificate is inactive. The implementation should increase the anchor budget when a reliable anchor exists, redesign the prediction/gain parameters, or mark the target temporarily unreachable rather than supply an uncertified cooperative estimate. Merely reducing the neighbor budget does not improve the direct-anchor factor $L_{x,j}(1 - w_{i0}^{(j)})$ when the anchor weight is unchanged.

C. Stability of the error

For target j , let $e_0^{(j)} = \hat{x}_0^{(j)} - x_j$, $e_i^{(j)} = \hat{x}_i^{(j)} - x_j$, and $e_{\mathcal{DO}}^{(j)} = \text{col}(e_1^{(j)}, \dots, e_N^{(j)})$. The local and distributed nonlinear error recursions induced by (??) and (??) are

$$e_0^{(j)}(t+1) = f_b(\hat{x}_0^{(j)}, u_j) + F_j(y_j - h_j(\hat{x}_0^{(j)})) - x_j(t+1), \quad (52)$$

$$e_i^{(j)}(t+1) = f_b(\hat{x}_{i,c}^{(j)}, \hat{u}_j) - x_j(t+1), \quad i \in \mathcal{V}, \quad (53)$$

where $\hat{x}_{i,c}^{(j)}$ is the trust-weighted correction in (??); time arguments on its right-hand side are omitted for readability.

Assumption 2 (Compact operating region). *The trajectories considered in the analysis remain in a compact set $\Omega_x \times \Omega_u$ with bounded speed and inputs, $|\delta| \leq \delta_{\max} < \pi/2$, accepted packets inside the receiving-side admissible set (??), and a local yaw chart without wrap discontinuities.*

Under Assumption ??, there are finite $L_{x,j}, L_{u,j} > 0$ such that

$$\|f_b(\xi, \mu) - f_b(\zeta, \nu)\|_* \leq L_{x,j}\|\xi - \zeta\|_* + L_{u,j}\|\mu - \nu\|_* \quad (54)$$

for all $(\xi, \mu), (\zeta, \nu) \in \Omega_x \times \Omega_u$. Conservative Lipschitz bounds and the local gain are computed from the trajectory Jacobians

$$\begin{aligned} A_i(t) &= \left. \frac{\partial f_b}{\partial x} \right|_{(\bar{x}_i, \bar{u}_i)}, \quad B_i(t) = \left. \frac{\partial f_b}{\partial u} \right|_{(\bar{x}_i, \bar{u}_i)}, \\ C_i(t) &= \left. \frac{\partial h_i}{\partial x} \right|_{\bar{x}_i} = I_4. \end{aligned} \quad (55)$$

Throughout the distributed-error analysis, we use the block maximum norm

$$\|e_{\mathcal{DO}}^{(j)}(t)\|_{b,*} := \max_{i \in \mathcal{V}} \|e_i^{(j)}(t)\|_*. \quad (56)$$

The same block-norm convention is used for other stacked state vectors below. By Lemma ??, $M^{(j)}(t) = I_N - \tilde{\mathcal{L}}^{(j)}(t)$ is entrywise nonnegative and its i th row sum is $1 - w_{i0}^{(j)}(t)$. **Componentwise, its entries follow from the grounded Laplacian:**

$$M_{ik}^{(j)}(t) = \begin{cases} w_{is}^{(j)}(t), & k = i, \\ w_{ik}^{(j)}(t), & k \neq i, k \in \mathcal{N}_i, \\ 0, & \text{otherwise,} \end{cases} \quad (57)$$

where $w_{is}^{(j)}(t)$ is the implicit self-prediction weight. The i th row sum is therefore $w_{is}^{(j)}(t) + \sum_{k \in \mathcal{N}_i} w_{ik}^{(j)}(t) = 1 - w_{i0}^{(j)}(t)$. Nonnegativity allows the triangle inequality to pass through the weighted correction. For every host i ,

$$\begin{aligned} \left\| \sum_{k=1}^N M_{ik}^{(j)}(t) e_k^{(j)}(t) \right\|_* &\leq \sum_{k=1}^N M_{ik}^{(j)}(t) \|e_k^{(j)}(t)\|_* \\ &\leq (1 - w_{i0}^{(j)}(t)) \|e_{\mathcal{DO}}^{(j)}(t)\|_{b,*}. \end{aligned} \quad (58)$$

Taking the maximum over i yields the induced mixing norm $\|M^{(j)}(t)\|_\infty = \max_i (1 - w_{i0}^{(j)}(t))$ used in the contraction arguments.

Stacking the trust-weighted correction and using the non-negative row-substochastic weights gives

$$\begin{aligned} \|e_{e,\mathcal{DO}}^{(j)}(t)\|_{b,*} &\leq \|M^{(j)}(t)\|_\infty \|e_{\mathcal{DO}}^{(j)}(t)\|_{b,*} \\ &\quad + \|w_0^{(j)}(t)\|_\infty \|e_0^{(j)}(t)\|_* + \|r_{\text{chart}}^{(j)}(t)\|_{b,*}, \end{aligned} \quad (59)$$

where $M^{(j)} = I_N - \tilde{\mathcal{L}}^{(j)}$, $w_0^{(j)}$ stacks the anchor weights, and $r_{\text{chart}}^{(j)}$ is zero in the no-wrap Euclidean chart (or a bounded second-order retraction remainder).

The Jacobians in (??) provide the usual auxiliary linearized model for selecting F_j and checking conservative contraction constants, but the result below is stated for the nonlinear recursions (??)–(??).

If $a_0^{(j)}$ and $a_l^{(j)}$ corrupt the anchor and neighbor- l channels, respectively, the correction at host i contains

$$\xi_{\text{att},i}^{(j)}(t) = w_{i0}^{(j)}(t) a_0^{(j)}(t) + \sum_{l \in \mathcal{M}_i^{(j)}(t)} w_{il}^{(j)}(t) a_l^{(j)}(t). \quad (60)$$

Here $\mathcal{M}_i^{(j)}(t) \subseteq \mathcal{N}_i$ denotes the corrupted neighbor channels seen by host i for target j ; for unattacked, unavailable, or rejected channels the corresponding attack vector is zero. **Note that this symbol clashes with the mixing matrix $M^{(j)}$; renaming it to $\mathcal{K}_i^{(j)}(t)$ would remove the ambiguity.**

By (??), this perturbation enters the nonlinear error recursion as an additive ISS input bounded by

$$\|d_{\text{att},i}^{(j)}(t)\|_* \leq L_{x,j} \|\xi_{\text{att},i}^{(j)}(t)\|_*. \quad (61)$$

Stacking these row-wise terms gives the additive input $d_{\text{att}}^{(j)}$. Its size is modulated by the trust weights. If $\|a_0^{(j)}(t)\|_* \leq \bar{a}_0$ and $\|a_i^{(j)}(t)\|_* \leq \bar{a}_N$ on $\mathcal{M}_i^{(j)}(t)$, then Lemma ?? gives

$$\begin{aligned} \|d_{\text{att},i}^{(j)}(t)\|_* &\leq L_{x,j} \left(w_{i0}^{(j)}(t) \bar{a}_0 + \sum_{l \in \mathcal{M}_i^{(j)}(t)} w_{il}^{(j)}(t) \bar{a}_N \right) \\ &\leq L_{x,j} \max\{\bar{a}_0, \bar{a}_N\}. \end{aligned} \quad (62)$$

Thus $\|d_{\text{att}}^{(j)}(t)\|_{b,*} \leq \bar{d}_{\text{att}}$ for some finite $\bar{d}_{\text{att}}$; after rejection, the corrupted channel's direct term in (??) is zero.

For the nonlinear ISS proof, $d_0^{(j)}$ and $d_{\text{DO}}^{(j)}$ denote the corresponding lumped inputs caused by model uncertainty, input mismatch, chart remainders, and bounded approximation terms. We assume these inputs and the additive attack contribution are bounded.

Assumption 3. There are finite $\bar{d}_0$, $\bar{d}_{\text{DO}}$, and $\bar{d}_{\text{att}}$ such that, for every t , $\|d_0^{(j)}(t)\|_* \leq \bar{d}_0$, $\|d_{\text{DO}}^{(j)}(t)\|_{b,*} \leq \bar{d}_{\text{DO}}$, and $\|d_{\text{att}}^{(j)}(t)\|_{b,*} \leq \bar{d}_{\text{att}}$. Hence $\|\tilde{d}_{\text{DO}}^{(j)}(t)\|_{b,*} \leq \bar{d}_{\text{DO}} := \bar{d}_{\text{DO}} + \bar{d}_{\text{att}}$.

Lemma 4 (Local nonlinear observer ISS). Assume that, on the compact operating region $\Omega_x \times \Omega_u$, the disturbance-free local nonlinear error recursion associated with (??) is uniformly locally exponentially stable. Equivalently, there exist $c_\ell > 0$ and $\lambda_\ell \in (0, 1)$ such that the zero-input local error satisfies the corresponding exponential bound for all sufficiently small initial errors and all admissible trajectories in the region. Then there exist local radii $r_\ell, d_\ell > 0$ such that, if $\|e_0^{(j)}(0)\|_* \leq r_\ell$ and $\sup_t \|d_0^{(j)}(t)\|_* \leq d_\ell$, the perturbed recursion remains in a local invariant neighborhood contained in the admissible region and is ISS with respect to $d_0^{(j)}$. In particular, for some finite local input gain $\gamma_\ell > 0$,

$$\|e_0^{(j)}(t)\|_* \leq c_\ell \lambda_\ell^t \|e_0^{(j)}(0)\|_* + \gamma_\ell \bar{d}_0. \quad (63)$$

A sufficient local check is the uniform exponential stability of the LTV local-error recursion

$$z(t+1) = (A_j(t) - F_j(t)C_j(t))z(t), \quad (64)$$

understood as an LTV system and not as frozen-time Schur stability of each $A_j(t) - F_j(t)C_j(t)$. A checkable sufficient condition is the existence of a bounded symmetric matrix sequence $P_j(t) = P_j(t)^\top$ and scalars $0 < \underline{p} \leq \bar{p}$ and $q > 0$ such that, with $A_{F,j}(t) := A_j(t) - F_j(t)C_j(t)$,

$$\begin{aligned} \underline{p}I &\preceq P_j(t) \preceq \bar{p}I, \\ A_{F,j}(t)^\top P_j(t+1) A_{F,j}(t) - P_j(t) &\preceq -qI \end{aligned} \quad (65)$$

for all t , together with a sufficiently small invariant neighborhood on which the higher-order remainder of the nonlinear local-error map is uniformly quadratic.

Proof. Uniform local exponential stability of the smooth zero-input recursion admits a local Lyapunov function with uniform quadratic bounds and a strict one-step decrease. On a sufficiently small invariant neighborhood, the additive input changes this Lyapunov difference by a term proportional to $\|d_0^{(j)}\|$. Iterating the resulting comparison inequality and using Assumption ?? gives (??) for a finite γ_ℓ . The Lyapunov inequality (??) establishes uniform exponential stability of the LTV linearization; the uniformly quadratic higher-order remainder then gives uniform local exponential stability of the nonlinear zero-input recursion on a sufficiently small invariant neighborhood. ■

Lemma 5 (Nonlinear distributed-error contraction). Suppose that the block contraction condition (??) below holds. If the local-observer error is ISS and Assumption ?? holds, then the nonlinear distributed error (??) is ISS with respect to $e_0^{(j)}$ and $d_{\text{DO}}^{(j)}$. Under the additive attack model, the attacked distributed error is ISS with respect to $e_0^{(j)}$ and $\tilde{d}_{\text{DO}}^{(j)}$.

Proof. Use the componentwise error-size vector $E^{(j)}(t)$ defined in (??). Equations (??) and (??), together with the anchor and disturbance terms, give a componentwise recursion of the form

$$E^{(j)}(t+1) \leq L_{x,j}(t) M^{(j)}(t) E^{(j)}(t) + b^{(j)}(t), \quad (66)$$

where $\|b^{(j)}(t)\|_\infty$ is bounded by finite gains multiplying $\|e_0^{(j)}(t)\|_*$ and $\|\tilde{d}_{\text{DO}}^{(j)}(t)\|_{b,*}$. Define the block mixing product

$$\Phi_M^{(j)}(t, T_c) := M^{(j)}(t + T_c - 1) \cdots M^{(j)}(t). \quad (67)$$

Unrolling (??) over a block of length T_c and using (??) gives

$$\begin{aligned} \|E^{(j)}(t + T_c)\|_\infty &\leq G_{x,j}(T_c) \|\Phi_M^{(j)}(t, T_c)\|_\infty \|E^{(j)}(t)\|_\infty \\ &\quad + c_b \sup_{\tau \in [t, t+T_c-1]} \|e_0^{(j)}(\tau)\|_* \\ &\quad + c_b \sup_{\tau \in [t, t+T_c-1]} \|\tilde{d}_{\text{DO}}^{(j)}(\tau)\|_{b,*} \end{aligned} \quad (68)$$

for some finite $c_b > 0$. Condition (??) contracts the homogeneous term by at most $\alpha_c < 1$. Iterating (??) gives a geometric ISS bound at the block times, and the bounded one-step gains extend it to the intermediate samples. Lemma ?? and Assumption ?? bound the inputs. In the nominal case, $d_{\text{att}}^{(j)} = 0$ and $\tilde{d}_{\text{DO}}^{(j)} = d_{\text{DO}}^{(j)}$. ■

Combining Lemmas ?? and ?? gives the following result.

Theorem 1. Fix a target $j \in \mathcal{V}$. Consider the vehicle platoon (??) over the directed weighted communication network $\bar{\mathcal{G}}^{(j)}$, and let (??) be the corresponding nonlinear distributed observer for target j . Assume that the following conditions hold:

- 1) There exists a bounded gain $F_j(t)$ such that the local nonlinear observer error recursion (??) is uniformly locally exponentially stable on the compact region of Assumption ??. A sufficient check is the uniform exponential stability of the LTV recursion (??) along the considered bounded bicycle trajectory, for instance through the parameter-dependent Lyapunov inequality (??) stated in Lemma ??. Frozen-time eigenvalue

placement of $A_j(t) - F_j(t)C_j(t)$ alone does not imply this property in the time-varying case.

- 2) For all $i \in \mathcal{V}$, $w_{i0}^{(j)}(t) \geq 0$ and $w_{il}^{(j)}(t) \geq 0$ satisfy

$$\sum_{l \in \mathcal{N}_i \cup \{0\}} w_{il}^{(j)}(t) \leq 1. \quad (69)$$

- 3) There exist a window length $T_c \geq 1$ and a scalar $\alpha_c \in (0, 1)$ such that, uniformly over all admissible switches, the infinity norm induced by the block maximum norm (??) satisfies

$$G_{x,j}(T_c) \left\| M^{(j)}(t + T_c - 1) \cdots M^{(j)}(t) \right\|_{\infty} \leq \alpha_c < 1. \quad (70)$$

Here $G_{x,j}(T_c)$ is the separable interleaved block gain defined in Lemma ??; the conservative choice is $L_{x,j}^{T_c}$. A checkable sufficient condition for $T_c = 1$ is Lemma ??; for intermittent anchors or sparse accepted graphs, Assumption ?? can be enforced by a supervisor or checked on each completed window, and Lemma ?? gives the corresponding block check (??).

Then the original nonlinear error dynamics (??)–(??) are locally ISS on $\Omega_x \times \Omega_u$. Under the additive attack model, the attacked estimation error is also locally ISS. The bounds below hold for sufficiently small initial errors and bounded-input suprema for which the perturbed errors remain in the local invariant neighborhoods of Lemma ?? and Lemma ??. In particular, choose any

$$\max\{\alpha_c^{1/T_c}, \lambda_{\ell}\} < \lambda < 1. \quad (71)$$

There then exist constants $c_0, c_1 > 0$ such that

$$\|e_0^{(j)}(t)\| \leq c_0 \lambda^t \|e_0^{(j)}(0)\| + c_0 \bar{d}_0, \quad (72)$$

$$\begin{aligned} \|e_{\mathcal{DO}}^{(j)}(t)\| &\leq c_1 \lambda^t \left(\|e_{\mathcal{DO}}^{(j)}(0)\| + \|e_0^{(j)}(0)\| \right) \\ &\quad + c_1 \left(\bar{d}_0 + \bar{\bar{d}}_{\mathcal{DO}} \right). \end{aligned} \quad (73)$$

Proof. Condition 1 and Assumption ?? give the local-observer ISS bound by Lemma ?. Lemma ?? gives Condition 2 and the nonnegative row-substochastic structure used in (?). Condition 3 gives the distributed block contraction in Lemma ?. Substituting (??) into the block recursion (??) produces the convolution of the block rate α_c^{1/T_c} with the local rate λ_{ℓ} . For any λ satisfying (?), this convolution is bounded by a finite constant times λ^t , including when the two original rates coincide. The finite within-block, bounded-gain, and norm-equivalence constants are absorbed into c_0, c_1 , which gives the stated decay of both initial errors and the bounded disturbance terms. In the nominal case $d_{\text{att}}^{(j)} = 0$ and $\bar{\bar{d}}_{\mathcal{DO}} = \bar{d}_{\mathcal{DO}}$. ■

Remark 4. The result is local under Assumption ?. The Jacobians in (??) provide conservative design constants, while the LTV local-error check must be performed along the trajectory rather than through frozen-time eigenvalues alone. Dense accepted-anchor graphs use $T_c = 1$; sparse switching graphs use the ordered windows of Assumption ?. Rejection drives the malicious input gain to zero.

VII. CONTAMINATION ROLLBACK UNDER DELAYED TRUST DECISIONS

Trust values are computed from consistency windows and therefore may lag behind the first corrupted packet. During this detection delay, falsified data can enter the distributed observer before the corresponding neighbor is removed from the trusted set. To limit this transient contamination, each host vehicle can optionally maintain a finite rollback buffer and re-evaluate the most recent observer steps once a neighbor is flagged.

Each host performs rollback using the data stored in its own buffer. It can remove a rejected source only when data from that source appear directly in the buffer. For example, suppose an attacked vehicle l^* sends false data to vehicle m , and vehicle m then sends a contaminated estimate to vehicle i . If vehicle i has stored only the estimate from m , replaying its buffer cannot identify and remove the contribution of l^* .

Correcting this indirect contamination would require either provenance information, meaning a record of the upstream packets and weights used to form each stored estimate, or a coordinated replay in which m corrects its history before i replays its own data. Therefore, the procedure below directly corrects only rejected data contained in the host's own buffer.

Assume that a trust decision is made at time t after the update that produced the pre-reset estimate $\hat{x}_i^{(j)}(t^-)$ and before the reset-corrected estimate at time t is released. Thus, replay over $k = t - K, \dots, t - 1$ produces the replacement value for $\hat{x}_i^{(j)}(t)$. For a rollback horizon K , host vehicle i stores, for each observer $\mathcal{DO}_i^{(j)}$ and each time $k \in \{t - K, \dots, t - 1\}$, the pre-update state $\hat{x}_i^{(j)}(k)$, the received neighbor and anchor estimates, the original weights, the input estimate, and the corresponding validity metadata. Let $\mathcal{R}_i^{(j)}(t) \subseteq \mathcal{N}_i$ be the neighbor sources retrospectively rejected by the new decision, let $\chi_{i0,r}^{(j)}(k; t) \in \{0, 1\}$ denote retrospective acceptance of the archived anchor, and let $\hat{u}_{j,r}(k; t)$ be an archived or recomputed input estimate that remains accepted at replay time. If no accepted bounded input estimate is available, replay is not certified and the target is marked temporarily unavailable. The replay is executed as follows:

- 1) form the retrospective neighbor, anchor, and input acceptance decisions from the trust decision at time t ;
- 2) reset a temporary replay state to the stored state at the beginning of the buffer;
- 3) replay the observer from $t - K$ to $t - 1$ using only archived channels that remain accepted;
- 4) replace the current distributed estimate by the replayed value.

The replay can remove only contributions introduced during the stored interval. Full removal additionally requires the buffer-start state $\hat{x}_i^{(j)}(t - K)$ to be an unaffected checkpoint; if the first affected packet predates the buffer or the checkpoint is already contaminated, replay is only a bounded mitigation step.

Define the retrospectively masked weights

$$\begin{aligned} \bar{w}_{il}^{(j)}(k; t) &= w_{il}^{(j)}(k) \mathbf{1}_{\{l \notin \mathcal{R}_i^{(j)}(t)\}}, \quad l \in \mathcal{N}_i, \\ \bar{w}_{i0}^{(j)}(k; t) &= w_{i0}^{(j)}(k) \chi_{i0,r}^{(j)}(k; t). \end{aligned} \quad (74)$$

During replay, the corrected state x_r is used to recompute the innovation terms

$$\tilde{\eta}_{il}^{(j)}(k; x_r) = \bar{w}_{il}^{(j)}(k; t)(\hat{x}_l^{(j)}(k) - x_r), \quad l \in \mathcal{N}_i, \quad (75)$$

$$\tilde{\eta}_{i0}^{(j)}(k; x_r) = \bar{w}_{i0}^{(j)}(k; t)(\hat{x}_0^{(j)}(k) - x_r). \quad (76)$$

The host resets the temporary state to the buffer start,

$$x_r \leftarrow \hat{x}_i^{(j)}(t - K), \quad (77)$$

and replays the observer update for $k = t - K, \dots, t - 1$:

$$x_r \leftarrow f_b \left(x_r + \sum_{l \in \mathcal{N}_i} \tilde{\eta}_{il}^{(j)}(k; x_r) + \tilde{\eta}_{i0}^{(j)}(k; x_r), \hat{u}_{j,r}(k; t) \right). \quad (78)$$

Finally, the current distributed estimate is corrected by setting

$$\hat{x}_i^{(j)}(t) \leftarrow x_r. \quad (79)$$

Because the masked weights in (??) do not exceed the original weights, their sum remains at most one; the removed weight remains assigned to the replay state through the residual self coefficient, so replay does not require renormalization. The rollback memory cost is proportional to $K \deg(i)$ per observer and can be implemented with archived packets, weights, inputs, and validity metadata. The anchor innovation is retained only when its archived channel remains accepted.

Lemma 6 (Bounded rollback reset). Assume that the stored packets, masked weights, accepted inputs, replayed states, and pre-reset estimates remain in the compact admissible region used in Theorem ??, and that the rollback horizon K is finite. At a reset time τ_q , define the stacked reset $\zeta_{\mathcal{DO}}^{(j)}(\tau_q)$ by stacking

$$\zeta_i^{(j)}(\tau_q) = x_r - \hat{x}_i^{(j)}(\tau_q^-)$$

for the resetting hosts and zeros for the other hosts. Then $\|\zeta_{\mathcal{DO}}^{(j)}(\tau_q)\|_{b,} \leq \bar{\zeta}$ for some finite $\bar{\zeta}$. Suppose also that every post-reset error remains in the local invariant neighborhood of Theorem ??, and that $\tau_{q+1} - \tau_q \geq T_r$ for an integer $T_r \geq 1$. Let $C_r > 0$ and $\rho_r \in (0, 1)$ bound the reset-free homogeneous distributed-error transition, as supplied by the block contraction condition. If $\mathcal{B}_{\text{ISS}}^{(j)}(t)$ denotes the reset-free ISS bound of Theorem ??, including norm-equivalence constants, then*

$$\begin{aligned} \|e_{\mathcal{DO}}^{(j)}(t)\|_{b,*} &\leq \mathcal{B}_{\text{ISS}}^{(j)}(t) + C_r \sum_{\tau_q < t} \rho_r^{t-\tau_q} \|\zeta_{\mathcal{DO}}^{(j)}(\tau_q)\|_{b,*} \\ &\leq \mathcal{B}_{\text{ISS}}^{(j)}(t) + \frac{C_r \bar{\zeta}}{1 - \rho_r^{T_r}}. \end{aligned} \quad (80)$$

Thus the reset-augmented observer is locally practically ISS with respect to the ordinary disturbances and the rollback-reset sequence.

Proof. For fixed finite K , the replay map is a finite composition of bounded weighted corrections and bicycle predictions on a compact set. It is therefore bounded. Since the pre-reset estimate is also bounded in the same admissible region, the difference between the replayed state and the pre-reset estimate is bounded, which gives a finite $\bar{\zeta}$. Between reset times, the block contraction proof of Theorem ?? gives a

transition bound $C_r \rho_r^{t-s}$. Inserting each reset as an additive jump and iterating this bound gives the first inequality in (??). The integer separation of reset times bounds the resulting geometric series by $(1 - \rho_r^{T_r})^{-1}$, which gives the second inequality. The post-reset neighborhood condition ensures that the local estimates used in this iteration remain valid. ■

Lemma ?? proves boundedness of the reset-augmented error; it does not guarantee that every rollback decreases the estimation error. Error reduction depends on a sufficiently early unaffected checkpoint and on correct retrospective channel decisions.

A. Pose propagation and relative-measurement re-anchoring

Rollback mitigates short-term contamination caused by delayed trust decisions. In contrast, re-anchoring provides an independent pose reference that limits the long-term propagation of accumulated position offsets when a trusted anchor or accepted relative measurement is available. This distinction separates ordinary pose propagation from optional relative-measurement re-anchoring.

Let $p = [X, Y]^T$ and write the consensus-corrected state as $\hat{x}_{i,c}^{(j)} = \text{col}(\hat{p}_{i,c}^{(j)}, \hat{\psi}_{i,c}^{(j)}, \hat{v}_{i,c}^{(j)})$. The usual observer prediction propagates the corrected pose as

$$\hat{p}_i^{(j)}(t+1) = \hat{p}_{i,c}^{(j)}(t) + T_s \hat{v}_{i,c}^{(j)}(t) \begin{bmatrix} \cos \hat{\psi}_{i,c}^{(j)}(t) \\ \sin \hat{\psi}_{i,c}^{(j)}(t) \end{bmatrix}. \quad (81)$$

Equation (??) is a motion update, not a position reset. If $\hat{p}_{i,c}^{(j)}$ is biased by contaminated cooperative data, then clean velocity or heading only gives a cleaner motion vector from a biased starting point. The position offset remains.

Re-anchoring is an optional correction before prediction. If host vehicle i has an onboard relative measurement to target j , and this measurement passes the same validity and trust checks used in the local-trust layer, then the host can build an independent pose reference for j . With the measured relative position expressed in the global frame,

$$p_{\text{rel},i}^{(j)}(t) = \hat{p}_0^{(i)}(t) + r_{i,j}^{\text{meas}}(t), \quad (82)$$

where $\hat{p}_0^{(i)}$ is the host local-observer position and $r_{i,j}^{\text{meas}}$ is the measured relative position from i to j . This reference does not come from the received distributed estimate of target j ; it comes from host sensing and the host's own local pose.

The host may then blend the corrected base pose with this relative-measurement reference:

$$\begin{aligned} p_{b,i}^{(j)}(t) &= (1 - \rho_{i,\text{rel}}^{(j)}(t)) \hat{p}_{i,c}^{(j)}(t) + \rho_{i,\text{rel}}^{(j)}(t) p_{\text{rel},i}^{(j)}(t), \\ \rho_{i,\text{rel}}^{(j)}(t) &\in [0, 1], \end{aligned} \quad (83)$$

and predict from the resulting base pose,

$$\hat{p}_i^{(j)}(t+1) = p_{b,i}^{(j)}(t) + T_s \hat{v}_{i,c}^{(j)}(t) \begin{bmatrix} \cos \hat{\psi}_{i,c}^{(j)}(t) \\ \sin \hat{\psi}_{i,c}^{(j)}(t) \end{bmatrix}. \quad (84)$$

The scalar $\rho_{i,\text{rel}}^{(j)}(t)$ determines how strongly the relative measurement is used. When $\rho_{i,\text{rel}}^{(j)}(t) = 0$, the update reduces to the standard observer prediction. Values $0 < \rho_{i,\text{rel}}^{(j)}(t) < 1$

give soft re-anchoring, while $\rho_{i,\text{rel}}^{(j)}(t) = 1$ resets the position base to the relative-measurement pose. If the relative measurement is missing, rejected, or outside the sensing range, then $\rho_{i,\text{rel}}^{(j)}(t) = 0$, and the observer reduces to the trust-weighted prediction and rollback mechanism analyzed above. Thus, re-anchoring is not an additional proof assumption; it is a practical add-on used only when an independent onboard relative measurement is available and accepted.

VIII. SIMULATION AND PLATFORM RESULTS

This section reports three validation layers. The numerical campaign uses a five-vehicle platoon and compares local, global, and simultaneous local-global corruption to evaluate RMSE, trust response, source suppression, and the dominant attack cases. The QCar/LIMO validation then checks rollback under real packet timing and uses trajectories to show closed-loop path effects. Finally, the control check uses trust as a gate between CACC and ACC and verifies positive projected spacing during the attack interval.

A. Numerical Simulation Setup

The numerical simulation uses five vehicles with all-to-all V2V communication; each vehicle runs the observer, and small fixed parameter variations are treated as modeling uncertainty. The main parameters are $N = 5$, $T_s = 0.01$ s, nominal geometry $l_f = 1.11$ m and $l_r = 1.74$ m, giving $L_w = 2.85$ m, local-trust product fusion with internal powers $(\omega_p, \omega_v, \omega_\psi, \omega_a) = (2.0, 3.0, 1.0, 1.0)$ for distance/position, velocity, heading, and acceleration/input consistency, class-filter aging weight $\beta_{\text{trust}} = 0.45$ in the active single-rating update, class confidence $C_{\text{tr}} = 0.2$, $q = 5$ trust classes, heading base tolerance $\psi_{\text{th}} = 0.35$ rad, and trust threshold $\theta_{\text{min}} = 0.5$. The observer-weight module uses base gains $w_{0,\text{base}} = 0.4$ and $w_{\text{self},\text{base}} = 0.2$, with nominal neighbor budget 0.4 and per-neighbor cap 0.4; after trust and flag normalization these are adaptive weights rather than fixed lower bounds.

For the direct-anchor case of Theorem ??, the contraction check uses the scaled infinity norm $\|z\|_* = \|D_x z\|_\infty$, $D_x = \text{diag}(0.01, 0.01, 1, 0.2)$, to account for mixed state units. On $v \leq 33$ m/s and $|\delta| \leq 0.35$ rad, the induced one-step bicycle gain evaluates to $L_{x,\text{max}} \approx 1.007$. With accepted anchor mass $\underline{w}_0 = \eta_0 \theta_{\text{min}} = 0.20$, Lemma ?? gives $\alpha = L_{x,\text{max}}(1 - \underline{w}_0) \approx 1.007 \times 0.80 \approx 0.806 < 1$. If a target anchor is rejected, no reconstruction from communication alone is claimed; the implementation holds or marks the estimate as temporarily unreachable until an accepted anchored path returns.

B. Evaluation Metrics

The numerical evaluation reports attack-window estimation error, trust response, and attacker-source suppression. All RMSE values below are computed on the attack interval $\mathcal{T}_{\text{att}} = [10, 15]$ s for non-attacker target vehicles. For a reported component $q \in \{d, v, a\}$, where d is the

distance/longitudinal-position error, v is velocity, and a is acceleration/input, the target-level RMSE is

$$\text{RMSE}_{q,i} = \sqrt{\frac{1}{|\mathcal{T}_{\text{att}}|} \sum_{t \in \mathcal{T}_{\text{att}}} (q_i(t) - \hat{q}_i(t))^2}. \quad (85)$$

The raw combined RMSE used for ranking is

$$S_{\text{raw},i} = \text{RMSE}_{d,i} + \text{RMSE}_{v,i} + \text{RMSE}_{a,i}. \quad (86)$$

Since this score adds quantities with different units, it is used only as a within-campaign severity index. Detection time is the first time at which the mean trust assigned to the attacker falls below $\theta_{\text{op}} = 0.70$; the corresponding detection delay is this time minus the 10 s attack onset. The average trust drop is the pre-attack attacker-trust mean minus the attack-window attacker-trust mean; larger values indicate a stronger reduction in the attacker's assigned credibility. The attacker-source zero-rate is the percentage of attack-window source-weight samples assigned to the attacker that are numerically zero, excluding direct self-weight.

C. Attack Scenarios

The numerical campaign uses a single corrupted source in a five-vehicle platoon. The corrupted vehicle is V1, and V2–V5 are the non-attacker targets used in the RMSE summaries. Each attack is active on $[10, 15]$ s. Table ?? lists the five attack cases. The same cases are evaluated under local, global, and simultaneous local-global corruption, so the channel comparison is made under matched attack definitions.

TABLE II
FIVE BICYCLE-STATE ATTACK IDS USED IN THE NUMERICAL CAMPAIGN.

Case	Type	Target	Parameters
1	Bias	Position X	bias = -5 m
2	Faulty	Position X	std. = 10 m, $p = 0.5$
3	Bias	Speed v	bias = -2.5 m/s
4	Faulty	Speed v	std. = 2.5 m/s, $p = 0.5$
5	Drop	All state packets	$p_{\text{drop}} = 0.5$

D. Numerical Simulation Results

All attacks are evaluated over the 10–15 s attack window using the raw combined RMSE S_{raw} . Table ?? compares the three corruption modes. Global corruption gives the lowest mean raw combined RMSE, 0.393, and the lowest dispersion, with standard deviation 0.069. Local corruption gives the highest average degradation, with mean raw combined RMSE 0.643. Simultaneous local-global corruption gives an intermediate mean RMSE of 0.567, but it creates the most severe single residual, $S_{\text{raw}} = 3.856$ on V5 under Case 2. The trust-drop column measures how much the mean trust assigned to the corrupted source decreases from the pre-attack baseline to the attack-window average; larger values indicate stronger trust penalization. The source zero-rate is a suppression measure: it is the fraction of attacker-source fusion weights that are set to zero during the attack window, excluding direct self-weight. Thus, a larger source zero-rate means the corrupted source is more often completely removed from distributed fusion.

TABLE III
AGGREGATE ATTACK-WINDOW PERFORMANCE BY CORRUPTION MODE.

Mode	Mean S_{raw}	Std. dev.	Max. S_{raw}	Trust drop	Source zero	Detect. time
Local	0.643	0.586	2.058	0.319	80.64%	10.088 s
Global	0.393	0.069	0.546	0.460	92.90%	10.028 s
Both	0.567	0.796	3.856	0.544	98.44%	10.030 s

Note: Std. dev. denotes standard deviation; trust drop is pre-attack attacker trust minus attack-window attacker trust; source zero denotes the attacker-source zero-rate; detection time is the mean first-threshold-crossing time.

TABLE IV
ATTACK-CASE SEVERITY RANKING BY MEAN RAW COMBINED RMSE.

Rank	Case	Mean	Attack	Local	Global	Both
1	2	1.232	X fault, 10 m	1.775	0.515	1.406
2	4	0.387	v fault, 2.5 m/s	0.361	0.405	0.396
3	3	0.363	v bias, -2.5 m/s	0.364	0.362	0.362
4	5	0.354	DoS	0.380	0.347	0.335
5	1	0.334	X bias, -5 m	0.334	0.334	0.334

Table ?? ranks the attack cases by the mean raw combined RMSE averaged over the three corruption modes. Case 2, the intermittent 10 m X fault, is the dominant attack: its cross-mode mean is 1.232, and its RMSE reaches 1.775 for local corruption, 0.515 for global corruption, and 1.406 for simultaneous corruption. The other four cases stay below 0.405 RMSE in all three configurations.

The quantitative comparison shows three clear patterns. First, global corruption reduces mean RMSE by 38.9% relative to local corruption. Second, simultaneous local–global corruption is 11.8% lower than local corruption but 44.2% higher than global corruption. Third, simultaneous corruption creates the largest single RMSE, 3.856, which is 87.4% higher than the local maximum and 7.06 times the global maximum. Detection reaches 100% in all cases and configurations. The average detection times are close to the 10 s attack onset, except local Case 4, where the mean detection time is delayed to 10.263 s. Therefore, local corruption causes the largest average estimation error, global corruption is least damaging on average, and simultaneous local–global corruption creates the highest worst-case outlier risk.

E. QCar/LIMO Platform Rollback and Closed-Loop Trajectory Diagnostics

1) *Platform and Communication Stack*: The observer is also implemented on a mixed QCar/LIMO 1:10 scale platform, where each vehicle runs the same estimation, communication, and control stack under real sensor timing, packet exchange, and controller execution. Vehicles broadcast local estimates, fleet estimates, trust summaries, and packet status. The receiving host checks physical and temporal consistency, updates trust weights, and runs the distributed observer with the local observer as an anchor. The numerical campaign uses labels V1–V5, whereas platform logs use V0, V1, and V2.

Figure ?? summarizes this flow: peer vehicles exchange UDP packets, the trust observer converts consistency checks

into adaptive fusion weights, and the fleet estimate is used for validation and control. The rollback comparison uses the existing cooperative controller; the spacing check uses the trust-blended cooperative/local implementation. The same attack channel supports matched no-rollback and rollback runs.

At the vehicle level, Fig. ?? separates longitudinal and lateral control while using the same planar bicycle state. The longitudinal branch projects the estimated position onto the road tangent and applies cooperative cruise control, so contaminated distributed estimates can affect the command before trust rejection and rollback correction. Trust weighting and rollback reduce this effect but do not guarantee complete rejection for every realization. The lateral branch uses cross-track and heading errors to compute steering. Both QCar and LIMO vehicles therefore share the normalized (X, Y, ψ, v) interface used for position, velocity, yaw, trust, and rollback-correction histories.

2) *Rollback Evaluation Protocol*: The platform validation focuses on delayed-decision rollback rather than deterministic RMSE ranking because packet delays, asynchronous sensing, and execution jitter can let corrupted packets enter the observer before trust drops below the rejection threshold. The primary comparison is the same attack configuration without rollback and with finite-window rollback. After delayed detection, the host replays the recent observer window with the corrected trusted-neighbor set. This is mitigation, not perfect attack rejection, because the CACC-only controller may already have reacted to contaminated states.

Diagnostic clean-motion tests are separated from deployable observer operation. Clean-motion replay can use uncorrupted heading and speed in (??), but it keeps the estimator's current base position and exposes residual pose contamination. A hard re-anchor through (??)–(??) is realistic only with an accepted onboard relative measurement or another independently validated pose source; re-anchoring to a clean reference that bypasses the attack injector is used only as an ablation/debugging mode. The QCar/LIMO code, resources, and rollback video are provided in the online resources listed in the author footnote.

3) *Rollback and Anchoring Results*: To isolate rollback and anchoring, Fig. ?? compares three runs under the same attack configuration: trust-based estimation without rollback, with rollback, and with rollback plus relative pose anchoring. Figure ?? illustrates the response of the trust-based distributed observer when the information used to estimate target vehicle V_0 is compromised. During the attack interval, the observer without rollback accumulates corrupted information, producing a position-estimation error of approximately 0.8 m and a pronounced deviation in the estimated velocity. The direct attacked-source weight decreases toward zero, indicating that the trust mechanism detects and suppresses the contribution of the compromised source. This behavior is also reflected in the reduction of the generalized trust value during the attack.

Although source rejection prevents further propagation of malicious information, it does not by itself remove the corrupted state already accumulated by the observer. The rollback mechanism addresses this limitation by replaying the buffered updates from a stored checkpoint. As shown in Fig. ??(e), rollback reduces the position error by 54.4% at the correction

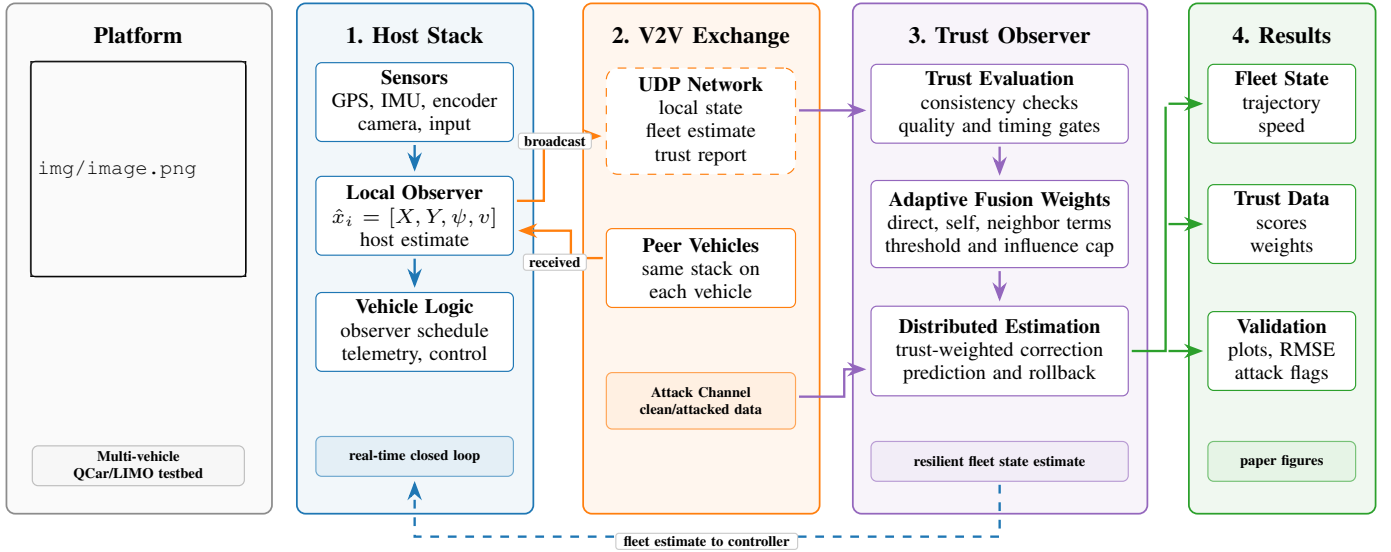


Fig. 2. QCar/LIMO experimental scheme for trust-aware cooperative state estimation. Each vehicle runs a local observer and vehicle-logic loop, exchanges V2V estimates and trust reports, and computes trust-weighted distributed estimates with optional rollback.

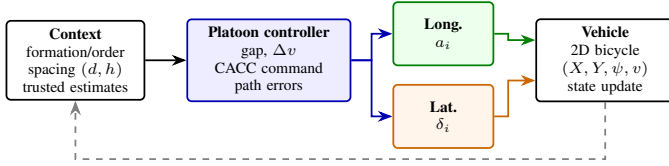


Fig. 3. Compact QCar/LIMO vehicle-level controller used in the rollback validation. The commands a_i and δ_i drive the 2D kinematic bicycle model.



Fig. 4. Case 3 local attack on QCar/LIMO: rollback and relative-pose anchoring under real-vehicle delay.

instant. Introducing anchoring further constrains the reconstructed estimate using accepted relative-pose information, re-

sulting in a 75.2% error reduction and maintaining the estimation error close to the pre-attack level. In this experiment, the three mechanisms perform complementary functions: rejection suppresses future direct influence from the flagged source, rollback revises its buffered direct contributions, and anchoring improves the consistency of the recovered estimate.

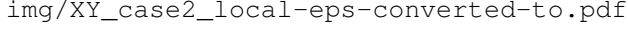
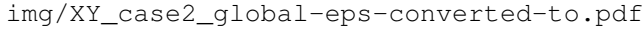
4) *Closed-Loop Trajectory Diagnostics*: The same platform logs were also replayed as XY trajectory diagnostics. Figures ?? and ?? plot the target- V_0 reference, pre-correction estimate, and final estimate for Case 2 local- and global-channel attacks. These plots quantify reconstruction quality on trajectories recorded during closed-loop operation; by themselves they are estimator/trajectory diagnostics rather than controller-safety evidence.

In both Case 2 trajectory diagnostics, rollback moves the final estimate closer to the reference, and rollback plus anchoring gives the smallest trajectory RMSE. The plotted RMSE decreases from 0.983 to 0.247 to 0.042 m in the local-channel run and from 0.195 to 0.141 to 0.023 m in the global-channel run. The trust and source-suppression behavior for the broader numerical campaign is summarized separately in Table ??.

F. Distributed State Estimation for Platoon Control

The trust scores are used both in the observer, where they limit unreliable V2V influence, and in the controller, where they gate cooperative CACC. Because a corrupted distributed estimate may affect the command before delayed trust correction, the longitudinal controller combines a local ACC channel based on direct sensing of the immediate predecessor with a cooperative CACC channel based on distributed platoon estimates. A trust gate uses CACC when the fleet estimate is credible and shifts toward ACC when it is not.

The planar bicycle states are first projected onto the road


Fig. 5. Case 2 local-channel XY trajectory reconstruction for target V_0 .

Fig. 6. Case 2 global-channel XY trajectory reconstruction for target V_0 .

tangent,

$$e_{\parallel}(k) = [\cos \psi_{\text{ref}}(k), \sin \psi_{\text{ref}}(k)]^{\top}, \quad (87)$$

$$s_i(k) = p_i(k)^{\top} e_{\parallel}(k), \quad (88)$$

where $p_i = [X_i, Y_i]^{\top}$. For follower $i \geq 2$, let $\mathcal{P}_i = \{1, \dots, i-1\}$ denote the cooperative predecessors. The equations use one-based platoon indexing; the experimental table keeps platform labels, where V_0 corresponds to vehicle 1, V_1 to vehicle 2, and so on.

TABLE V
ATTACK-WINDOW CLOSED-LOOP METRICS UNDER THE PLATFORM V_0
CASE 2 ATTACK.

Host	Δt_{det} [s]	$\bar{\gamma}_{i,i-1}^{\text{att}}$	$\bar{w}_{\text{src}}(V_0)$	$s_{\min}$ [m]	Gap RMSE [m]
V1	0.093	0.227	0.041	0.681	0.345
V2	0.090	0.894	0.048	0.637	0.171

Note: $\bar{\gamma}_{i,i-1}^{\text{att}}$ is the mean direct-predecessor CACC fusion gate during attacks; $\bar{w}_{\text{src}}(V_0)$ is the mean observer source weight assigned to the attacked V_0 channel.

The closed-loop validation uses a platform V_0 attack schedule containing the dominant Case 2 intermittent position-fault family; the injected timeline is shown in Fig. ??(d). This is a check of the dominant attack family, not a repeat of the exact worst attacker–target cell. Safety is evaluated by the minimum projected bumper-to-bumper spacing

$$s_{\min} = \min_{k \in \mathcal{T}_{\text{att}}} \min_{i=2, \dots, N} ((p_{i-1}(k) - p_i(k))^{\top} e_{\parallel}(k) - L). \quad (89)$$

The tested interval is longitudinally collision-free when $s_{\min} > 0$; the platform table uses 0.30 m as an experimental safety margin. This path-projected check does not prove full two-dimensional collision avoidance when lateral deviations are large.

The local adaptive-cruise branch uses only the measured predecessor gap and the time-headway target $d_{\text{ACC}} + h_{\text{ACC}} \max\{\hat{v}_0^{(i)}, 0\}$, so it remains available when vehicle-to-vehicle data are rejected. The cooperative branch uses distributed predecessor position, speed, and heading errors:

$$u_{\text{coop},i}(k) = \sum_{j \in \mathcal{P}_i} [\kappa_s e_{s,ij}(k) + \kappa_v e_{v,ij}(k) + \kappa_{\psi} e_{\psi,ij}(k)], \quad (90)$$

where $e_{s,ij}$, $e_{v,ij}$, and $e_{\psi,ij}$ are the corresponding relative errors and $d_{i,j} = (i-j)(d_{\text{ACC}} + h_{\text{ACC}} \hat{v}_0^{(i)})$ for a homogeneous platoon under the same spacing policy. The trust-fused, filtered command is

$$\begin{aligned} u_{\text{target},i} &= u_{\text{loc},i} + \gamma_i(u_{\text{coop},i} - u_{\text{loc},i}), \\ \gamma_i &= \mathcal{VT}_{i,i-1}, \\ u_i(k) &= u_i(k-1) + \tau_f(u_{\text{target},i}(k) - u_i(k-1)). \end{aligned} \quad (91)$$

Invalid or unavailable predecessor packets set $\gamma_i = 0$, selecting the local branch; credible data move the command toward cooperative control. Non-immediate corrupted sources are already attenuated by the observer weights. The experiment tests attack-window projected spacing, not string stability.

Table ?? summarizes two follower-host logs under the same V_0 attack schedule. In both, the attacked V_0 source weight falls below 0.05 on average within about 0.1 s. V_1 shifts strongly toward local ACC because it directly follows V_0 , whereas V_2 keeps more cooperation because its direct predecessor V_1 remains consistent and V_0 is already suppressed at the observer input. Both hosts preserve positive projected spacing: the minimum gaps are 0.681 m and 0.637 m, above the 0.30 m margin. Thus, the reported gap RMSE reflects imperfect CACC tracking under attack, not loss of longitudinal collision clearance.

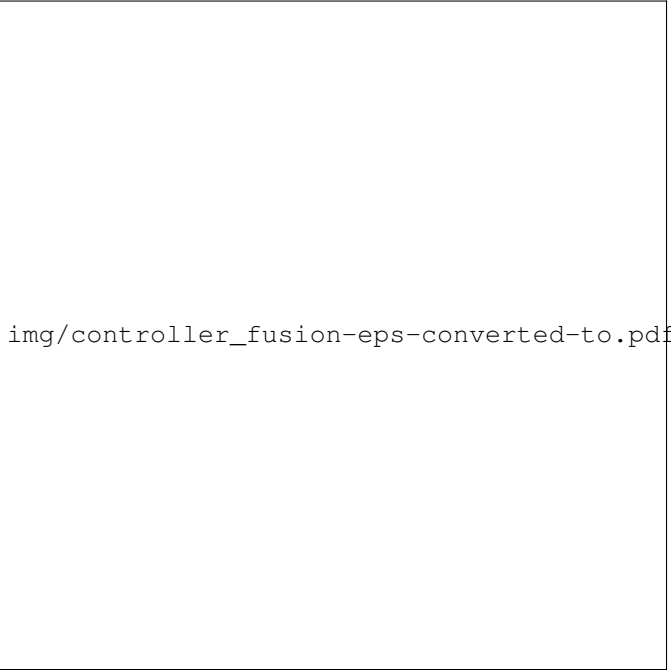


Fig. 7. Trust-gated ACC/CACC response and projected spacing under the platform V_0 attack schedule.

IX. DISCUSSION AND CONCLUSION

This paper developed a trust-aware distributed observer for resilient state estimation in connected vehicle platoons. The design couples three elements: a local-observer anchor that prevents consensus from drifting to a common biased value, trust-adaptive fusion weights that reduce the influence of inconsistent neighbors without discarding cooperation that remains physically plausible, and an optional finite-window rollback that revises recent updates once a delayed trust decision arrives. The analysis gives sufficient conditions for local nonlinear ISS on a compact operating region, covering bounded model uncertainty, input mismatch, accepted corruption, and the switching induced by the trust layer. Two checks are provided: a direct one-step condition when the target anchor is accepted, and a windowed condition over sequentially rooted accepted graphs when anchors are intermittent. Across the numerical campaign, the trust layer detected every injected attack and drove the corrupted source weight to zero in most attack-window samples. On the QCar/LIMO platform, rollback recovered a large part of the position error introduced during the detection delay, and the trust-modulated ACC/CACC controller preserved positive path-projected spacing throughout the tested attack interval.

The guarantee is deliberately conditional, and its boundaries are worth stating precisely. The receiving-side gate (??) projects every accepted packet onto a physically admissible set, so an arbitrarily large injection is either rejected or converted into a bounded input entering Theorem ?? through $\bar{d}_{\text{att}}$. What the theorem therefore certifies is graceful degradation under a specified admissible attack bound, not universal detection. The adversary that this formulation does not exclude is the patient one: a slowly ramping, physically consistent bias

that stays inside Ω_z and below the trust-rejection threshold will be accepted, and it can leave a nonzero ultimate estimation error. Characterizing the largest $\bar{d}_{\text{att}}$ reachable by such a stealthy strategy is the sharpest open question raised by this work.

Two further boundaries follow from the architecture itself. First, the method protects the platoon estimate rather than reconstructing a compromised vehicle: once a neighbor is rejected, recovering its true state requires an independent trusted measurement, a reliable local anchor, or a separate attack-identification module, and if an attack removes every reliable path from an accepted anchor to a target, communication alone cannot reconstruct that target. Second, rollback is a bounded mitigation, not a repair: it corrects only contributions held in the finite buffer, it presumes an uncontaminated checkpoint at the buffer start, and its ISS bound relies on a dwell time between resets.

The closed-loop evidence should be read with the same care. The experiment establishes that path-projected inter-vehicle spacing remains positive for the reported attack schedule. It does not establish string stability, formal f -resilience under a bounded fraction of adversaries, or full two-dimensional collision avoidance when lateral deviations are large. These are properties of a different order, and the present controller check does not reach them.

Future work follows directly from these boundaries. The first priority is the stealthy regime: quantifying the steady-state error produced by near-threshold ramps, and reporting benign false-positive rates under realistic sensor and communication noise, since the value of a trust gate is set as much by how rarely it rejects honest neighbors as by how quickly it rejects attackers. A second priority is to strengthen the evaluation itself, replacing the unit-inconsistent combined RMSE by a scaled-norm criterion in the same D_x metric used for the contraction check, and adding post-attack recovery. Beyond this, the natural extensions are matrix-valued trust weights for channel-wise attack isolation, tighter dwell-time and contraction conditions for rollback-triggered resets, sparse and dynamic topologies under coordinated attacks, a string-stability analysis of the trust-modulated closed loop, and validation on larger QCar/LIMO fleets.